

Chapter 19

A Detailed Study of Semiconductor Wafer Drying

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19.1 INTRODUCTION

19.1.1 Scope

In almost every industry, manufacturing steps can be found where a solid is brought in contact with a processing liquid. The function of this processing liquid is to act as a medium that transports dissolved or suspended species toward the surface of the solid (for example, in coating or electroplating applications) or away from the surface of the solid (for example, in etching or cleaning applications). In most cases, when the contact between solid and liquid is stopped, an amount of residual processing liquid remains attached to the surface. This liquid is removed in a rinsing and subsequent drying step. In the rinsing step, the residual processing liquid is diluted in the rinsing liquid (which in most cases is either water or a dedicated solvent). The remainder of this rinsing liquid, which has now replaced the initial processing liquid, is removed in the drying step.

Because the rinsing and drying steps are perceived as not contributing actively to the manufacturing process, they are often not well investigated or optimized. However, a badly chosen rinsing or drying step can result in the formation of residues. These are unwanted deposits of soluble or suspended contaminants (such as very small particles) on the solid substrate. The presence of these residues can degrade the overall process performance or reduce the surface quality. The sources of these suspended or soluble contaminants can be (1) the processing hardware (for example, particles formed due to friction between moving parts), (2) the rinsing liquid (e.g. impurities present in the source of rinsing liquid), and (3) the initial processing liquid (when it is not sufficiently removed during rinsing and drying). Depending on the particular process conditions each of these sources can be the dominant one. It is important to note, however, that the cause of the first two sources is mostly hardware related. Their importance can be readily reduced by increasing the purity of the rinsing liquid and by improving the overall tool particle neutrality. The latter case is of a different nature and requires a thorough optimization of the rinsing and drying steps. In order to do so, a thorough understanding of the transport of contaminants throughout the rinsing and drying steps is needed. In the literature, it has already been shown how particles removed from the surface by a wet cleaning step could re-adsorb on the surface due to pH transients during the rinsing step.¹

In this chapter, we shall focus on the drying step. Emphasis will be placed on drying techniques used in semiconductor device manufacturing which is, due to its very stringent surface cleanliness specifications, very sensitive to many types of surface contamination. For example, it has already been shown that incomplete removal of liquid during the drying step can result in *drying marks* that can degrade process performance^{2–5} either by deposition of dissolved contaminants or re-growth of a chemical oxide on HF-last surfaces.⁶

19.1.2 Approach Followed in this Work

In this work, the performance of several wafer drying techniques is compared using salt residue tests.⁷ In these tests, wafers are brought into contact with a solution containing tracer elements (metal salts, concentration C_0) and dried. The final surface concentration of tracer elements, σ , is determined after the drying step. For a correct interpretation of the results, one must realize that the metal salts can end up on the surface in two ways^{1,8}:

- *Adsorption during contact with the liquid.* As long as the wafer is in contact with the liquid, the tracer elements (dissolved or suspended species) can adsorb onto the surface. This contribution will be denoted by $\sigma_{\text{adsorption}}(t)$.
- *Deposition due to evaporation during drying.* During the drying step liquid can be removed from the wafer surface in two ways.⁹ In the first approach (referred to as convective removal), the liquid is removed in its liquid state. Together with this liquid all dissolved and suspended contaminants are also removed from the wafer surface. Alternatively, however, the liquid can be removed by evaporation. When this happens, all non-volatile dissolved and suspended contaminants in this evaporating liquid deposit on the surface of the solid. This contribution will be denoted by $\sigma_{\text{evaporation}}(t)$ and will be referred to as *evaporative deposition*.

Hence the surface concentration of tracer elements after drying can be written as:

$$\sigma(t) = \sigma_{\text{adsorption}}(t) + \sigma_{\text{evaporation}}(t) \quad (19.1)$$

where t denotes the processing (= immersion + drying) time. The volume of liquid that evaporates during the drying step (per unit area) can be expressed by an equivalent *evaporated thickness*.⁹ If we denote the average concentration of tracer elements in this evaporated volume by $\bar{C}(t)$, then we can write the contribution of evaporation to the total surface concentration of species as:

$$\sigma_{\text{evaporation}}(t) = \bar{C}(t) \times \delta_{\text{evap}} \quad (19.2)$$

where we assume the evaporated thickness δ_{evap} to be a constant that is specific for a given set of drying conditions. We will show later that it can also be used as a figure of merit to compare the performance of various drying techniques. Because the average concentration of tracer elements inside the evaporated volume can vary with time—for example, as a result of local depletion due to

adsorption onto the surface—we include the time t explicitly in the equation. By combining Eqs. (19.1) and (19.2), we obtain:

$$\sigma(t) = \sigma_{\text{adsorption}}(t) + \bar{C}(t) \times \delta_{\text{evap}} \quad (19.3)$$

where all terms are expressed as a surface concentration (i.e., species per area). By measuring the surface concentration of tracer elements in carefully designed experimental conditions, the relative contributions of both terms on the right-hand side of Eq. (19.3) can be estimated. For example, when the experimental conditions are chosen such that $\sigma_{\text{adsorption}}(t)$ can be neglected compared to $\bar{C}(t) \times \delta_{\text{evap}}$ we can write:

$$\sigma(t) \simeq \bar{C}(t) \times \delta_{\text{evap}} (\sigma_{\text{adsorption}} \ll \sigma_{\text{evaporation}}) \quad (19.4)$$

Hence, the evaporated liquid film thickness can be calculated as:

$$\delta_{\text{evap}} \simeq \frac{\sigma(t)}{\bar{C}(t)} (\sigma_{\text{adsorption}} \ll \sigma_{\text{evaporation}}) \quad (19.5)$$

if the surface concentration $\sigma(t)$ and the average concentration in the liquid $\bar{C}(t)$ are known.

19.1.3 Drying Techniques Commonly Used in Semiconductor Manufacturing

In semiconductor processing, wafer drying is commonly performed on a spin dryer. However, due to performance limitations new techniques such as isopropyl alcohol (IPA) vapor dryers and tensioactive (surface active) vapor assisted (TAV) dryers have found widespread use in semiconductor cleaning.^{3,10} In this work, the performance of spin drying and TAV drying are compared. A brief description of both techniques is presented below.

19.1.3.1 Spin Drying

Spin drying or centrifugal drying under laminar air flow is the oldest technique used to dry semiconductor wafers. In this technique, the wafer is rapidly (on the order of a few 1000 rpm) rotating around an axis perpendicular to the wafer surface. The axis of rotation can be horizontal or vertical and can be in the center of the wafer or lie outside the wafer (for example, for tools where multiple wafers are rotating around one central axis through their center of gravity). In this work, only single wafer horizontal spin drying is investigated. A schematic representation of this process is given in Figure 19.1.

Spin drying has the advantage of being a relatively simple and fast technique. Disadvantages are the risk of back splash of droplets and increased evaporation when high spin speeds are used.^{11,12} Additionally, electrostatic charging of the wafer surface takes place due to friction between the fast air stream¹³ which can attract particles from the gas phase. The charging effect can be

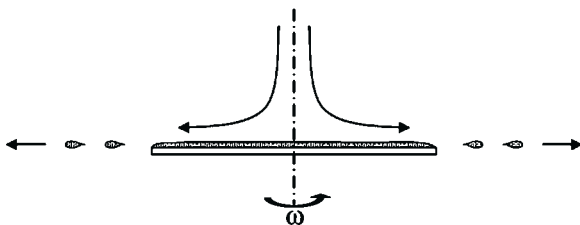


FIGURE 19.1 Schematic representation of the spin drying process on a single horizontally rotating wafer.

reduced by using ionizers,¹³ or adding carbon dioxide to the gas phase.^{14,15} It has been observed that drying techniques using IPA do not suffer from these charging effects.¹³

19.1.3.2 Surface Tension Gradient (Marangoni) Based Drying

Surface tension gradient-based drying techniques are also commonly referred to as *Marangoni*-based drying techniques as they make use of the Marangoni effect. For the sake of completeness, we shall first briefly explain this Marangoni effect. Next we shall introduce two surface tension gradient-based drying techniques investigated in this chapter.

19.1.3.2.1 The Marangoni Effect

The Marangoni effect was first described in the 1860s by the Italian physicist Carlo Giuseppe Matteo Marangoni who investigated the spreading of oil drops on a water surface.¹⁶ He explained this behavior as the macroscopic manifestation of a liquid flow as a result of local differences (gradients) in interfacial tension. Many other examples of manifestation of this phenomenon can be found in Refs.^{17–20} For liquids the presence of even small amounts of foreign species can significantly reduce the surface tension.^{21,22} This is especially true for water that has an intrinsically high surface tension of 72.8 mN/m at room temperature²³ due to its polar nature with many hydrogen bond possibilities. Because this value is significantly larger than for most other liquids (which are typically on the order of 20–40 mN/m), a large surface tension gradient exists due to a concentration gradient of foreign molecules at the water interface. This effect has been shown to cause a significant increase of the mass transfer in liquid–liquid extraction processes^{20,24} or cause spreading of liquids on water at speeds up to several tens of cm/s.^{25–31}

19.1.3.2.2 Vertical Marangoni-Based Drying of Silicon Wafers

A vertical drying technique for silicon wafers by means of the Marangoni effect has been described by Leenaars et al.³² using vapors from various types of tensioactive species such as alcohols, alkanes, ethers, and ketones.

They propose the following mechanism^{32–35}: at the contact region between liquid, solid, and vapor, a meniscus will be formed if the contact angle of the liquid is smaller than 90° . When a soluble volatile surface active component—present in the vapor phase—dissolves in the liquid, the concentration of the surface active component in the thinnest part of the liquid meniscus increases faster than in the bulk of the liquid. As a consequence, liquid flows from the thin part of the meniscus (having the lowest surface tension) toward the bulk of the liquid (having the highest surface tension) allowing the wafer to be withdrawn visually dry from the liquid bath. This principle is illustrated in Figure 19.2.

In semiconductor industry, this method is widely used because of its superior performance with respect to drying marks.^{2,3,10,34,36} In a typical industrial set-up, wafers are vertically withdrawn at a speed of approximately 1 mm/s from water into a vapor phase containing a solvent. This vapor is usually generated by saturating air or nitrogen with a solvent, by means of a bubbler system. The solvent used in these cases is almost exclusively IPA. The main reason is the fact that IPA is already widely used in semiconductor processing as a solvent and hence is readily available. Vertical IPA vapor-based drying is replacing the traditional IPA vapor dryers that work by solvent displacement when the wafer with a liquid layer is brought in an ambient of pure, heated IPA vapor. Besides an improved drying performance, additional benefits are obtained from this transition. Examples are a significant reduction in IPA usage and a

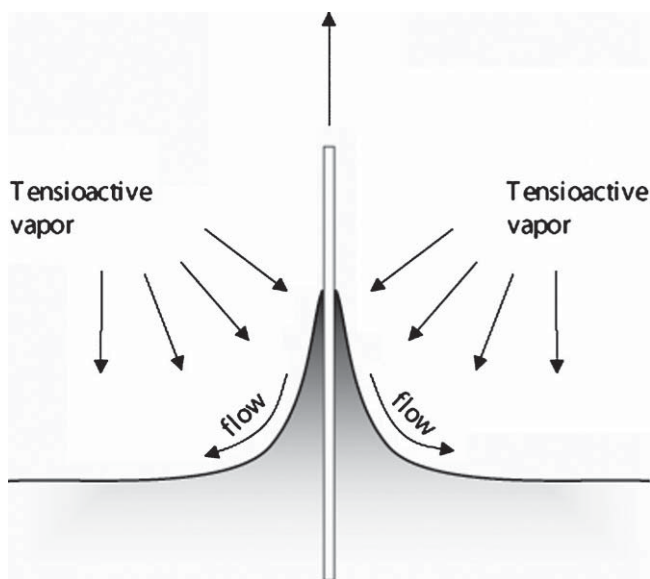


FIGURE 19.2 Schematic representation of the Marangoni effect in the meniscus of a liquid wetting a solid substrate.

reduction of safety problems associated with the risk of explosion^{2,3,10} when heated IPA is involved.

19.1.3.2.3 Marangoni-Based Drying of a Horizontally Rotating Wafer

Recently, a novel technique was proposed to perform a Marangoni-based drying process on a horizontally rotating wafer using two dispense nozzles (9, 37–40, see also Section 19.3.1.3 and Figure 19.11). In this method, the tensioactive vapor is delivered to the wafer surface through the first nozzle. At the same time, water is dispensed through a second nozzle that is located a few mm further away from the axis of rotation. As a result of this, a surface tension gradient in the liquid is induced in the region between the two nozzles. Because the wafer is rotating around its axis, a centrifugal force is induced that prevents the liquid from spreading radially inwards. The combination of the two adjacent flows of tensioactive vapor and water creates a meniscus region over which a concentration gradient develops. This induces a radially outward flow in the meniscus region. Drying of the wafer is accomplished by first positioning both nozzles over the center of the wafer and subsequently moving both nozzles at the same speed toward the edge of the wafer. The advantage of this technique is that the rotation speed is significantly lower than in the case of spin drying. This reduces the risk of adverse effects related to high speed spinning such as back splash of droplets and increased evaporation.^{11,12} Furthermore, the total distance over which the meniscus has to move is halved with respect to vertical Marangoni-based drying. In case the maximum allowed drying speed of the meniscus is comparable for both techniques, it will result in a decrease of the total drying time of at least a factor of two in the case of rotational Marangoni drying.

19.2 THEORETICAL BACKGROUND

The theoretical background needed for a correct interpretation of the salt residue tests (introduced in Section 19.1.2) is given here. It consists of some physical, chemical, and mathematical models that are relevant for the systems of interest. This starts with a study of the stability of thin liquid films on wettable substrates based on the theory of surface forces. Because most of the experimental data are obtained on O₃-grown hydrophilic silicon wafers (covered with hydrated SiO₂⁴¹), we restrict our review to wetting films on silica surfaces. Further, a model is discussed that can describe the adsorption of ions onto a silica surface.

19.2.1 Stability of Wetting Films on Silica

19.2.1.1 Surface Forces

On a microscopic scale, the transition between two bulk phases, for example, between a liquid and a solid, does not occur abruptly but rather over a zone of finite thickness, called the *transition zone* or the *interfacial region*.⁴² In this

region, properties of intensive parameters—such as pressure, concentration, density, etc.—differ from their bulk values. This gives rise to a field of electronic and molecular forces decreasing with distance into the bulk of the continuous phases. These forces are commonly denoted as *surface forces*⁴³ since they are due to the presence of a surface—or rather an interfacial region. Various excellent textbooks exist describing different types of surface forces and their origin.^{43,44} The most studied of these are the *electrostatic* and the *electrodynamic* (or van der Waals) forces. These are commonly referred to as DLVO forces, named after the scientists that first used these forces to describe the stability of solid/liquid systems (Derjaguin, Landau, Verwey, and Overbeek). Other forces (often referred to as *non-DLVO* forces) which are of importance for wetting systems on a silica surface are *short-range repulsive* (SRR) forces.^{1,8} The latter is a more general name for forces ascribed to a structural orientation of water molecules in the vicinity of the silica surface.^{1,43} The importance of these forces will be discussed more in detail in Section 19.2.1.3.

19.2.1.2 Disjoining Pressure

An important concept in the study of such wetting films is the *disjoining pressure*, introduced by Derjaguin in the 1930s.^{43,45,46} To illustrate this concept, consider a “thick” and a “thin” wetting film, as depicted in Figure 19.3. The hatched areas represent the transition zones at the top and bottom interfaces of the wetting film.

For the thick film (Figure 19.3(a)), there is no overlap between the transition zones. A change in thickness of such a film does not change the free energy of the system.⁴³ For the thin film (Figure 19.3(b)), the interfacial zones start to overlap, which increases the free energy of the system. As a result, the equilibrium hydrostatic pressure in a thin film in thermodynamic equilibrium will differ from the hydrostatic pressure in the bulk phase from which the film was formed by thinning down. The difference between these two pressures is defined as the disjoining pressure Π ,^{43,45,46} which represents the force per area that must be applied to maintain thermodynamic equilibrium in a thin film when it would be pressed between two solids. More generally, the disjoining pressure isotherm $\Pi(h)$ is defined as:

$$\Pi(h) = - \left(\frac{\partial G}{\partial h} \right)_{T, \mu_i, P} \quad (19.6)$$

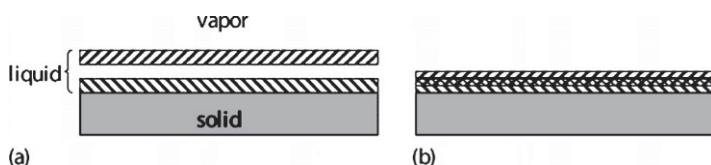


FIGURE 19.3 Schematic representation of (a) thick wetting film and (b) thin wetting film. Hatched areas represent the transition zones at solid/liquid and liquid/vapor interfaces.

with h the thickness of the film, G is the Gibbs free energy of the surface, with its derivative taken at constant temperature T , pressure P , and chemical potentials of all components μ_i . It has been shown that the disjoining pressure can also be written as a superposition of different components, each of them related to a corresponding surface force (e.g., dispersion, electrostatic, structural, adsorption, and electronic^{43,46}).

19.2.1.3 Importance of Short Range Repulsive Interactions

19.2.1.3.1 Aqueous Films

The investigation of aqueous films on various types of silica (silica powders, glass slides, quartz, silicates, etc.) has been the subject of numerous studies.^{43,46–52} As discussed above, the film stability is commonly described by a disjoining pressure isotherm. Experimentally such an isotherm can be determined from surface force apparatus measurements, by pressing a gas bubble against a flat plate or by interferometric measurements in a controlled ambient. A qualitative drawing of a disjoining pressure isotherm for water on quartz is given in Figure 19.4.^{43,44,46,47,53}

The x -axis represents the film thickness h , the y -axis represents the disjoining pressure Π and the relative vapor pressure in the gas phase p/p_{sat} . Both are related by Eq. (19.7)^{43,50,54}:

$$\Pi = -\frac{R_g T}{V_m} \ln \frac{p}{p_{\text{sat}}} \quad (19.7)$$

with V_m the molar gas volume and R_g the ideal gas constant. It can be seen that the isotherm follows an S-shaped profile that extends into the region of negative disjoining pressures (i.e., in the region of supersaturation). However, from thermodynamic considerations a film can only be stable if $(\partial \Pi / \partial h) < 0$ (see Eq. (19.6)).

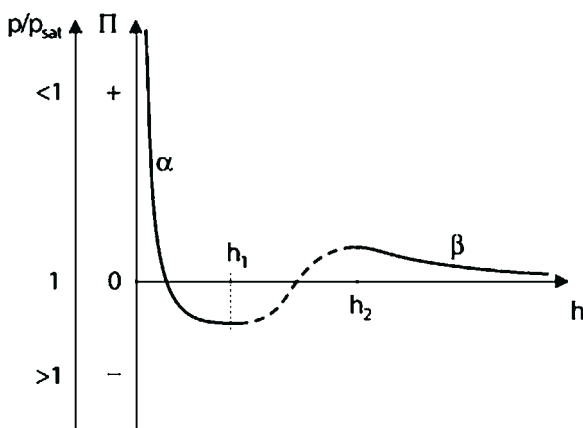


FIGURE 19.4 Schematic drawing of a disjoining pressure isotherm of water on quartz.

In Figure 19.4, we can, therefore, distinguish two stable branches. The region where $h < h_1$ is commonly denoted as the α -branch, the region where $h > h_2$ is referred to as the β -branch.^{43,50,54} Values for the thickness are typically below 10 nm for α -films and in the range of several tens of nm (up to 200 nm) for β -films.

The β -films are mainly stabilized by electrostatic forces, which are usually dominant for films thicker than 50 nm. In most electrolyte solutions, both interfaces ($\text{SiO}_2/\text{H}_2\text{O}$ and $\text{H}_2\text{O}/\text{air}$) are strongly negatively charged (e.g. -155 and -55 mV, respectively, in 10^{-5} M KCl⁴⁸) leading to a strong repulsive contribution to the disjoining pressure. The asymmetry of the film (with different values of the surface potentials) is also responsible for the S-shape of the curve. At short distances, the electrostatic contribution will then become attractive.⁵⁵ The relation between β -films and electrostatic forces is evidenced by the strong dependence of β -isotherms on electrolyte concentration.⁵⁰

For thinner films ($h < 50$ nm), van der Waals forces start to become important. Due to the spectral properties of water and silica, the Hamaker constant for a water film on SiO_2 is negative ($\approx -1 \times 10^{-20}$ J^{44,47,50,56}). This means that the van der Waals interactions also contribute to the stability of thin water films on silica surfaces. Furthermore, it has also been confirmed that experimental results on thin α -films deviate from the models based on electrostatic and van der Waals interactions only. The deviation has been ascribed to the presence of a structural force.^{43,46–52} It is believed that this structural force is related to the orientation of polar liquid molecules around a charged interface and is strongly repulsive at distances up to 10 nm. Because only electrostatic and van der Waals forces have been studied extensively, the structural force is often calculated from the deviation of experimental values from theoretical predictions of the first two components.

19.2.1.3.2 Alcohol Films

Similar to the case of aqueous films, there has also been evidence of short-range repulsive forces for alcohol films.⁵⁴ Derjaguin and Zorin have published an extensive study of the surface condensation of various types of alcohols on a glass surface.⁵⁴ For ethanol, propanol, and butanol, they found similar thicknesses for the α -films as for water vapor (up to about 6 nm at saturation). This finding was attributed to the presence of structural forces related to the polarity of the alcohol molecules. This means that for water/alcohol mixtures, surface forces can result in a stable thin wetting film with a thickness of a few nm. The same finding was also proposed by Marra and Huethorst³³ when studying the contraction of droplets in an ambient containing a tensioactive vapor.

19.2.2 Adsorption of Ions on Silica Surfaces

As discussed in Section 19.2, metal salts can be used as tracer elements to investigate the drying step. These tests are performed on hydrophilic silicon wafer

substrates (that is, wafer surfaces covered with a thin (<1 nm thick) chemical silicon dioxide layer^{57,58}). For a correct interpretation of the experimental results, it is necessary to have a representative model that can describe the interaction between metal ions and a silica surface in a solution. First, we present a model describing the chemical nature of a silica surface in contact with water. Next, we discuss how the metal ions in solution interact with this surface.

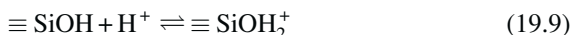
19.2.2.1 Structure of the Silica–Water Interface

The study of the properties of silica (SiO_2) has been the objective of numerous papers and books.^{59,60} Especially in the 1950s–1970s, finely dispersed and porous silicas became a subject of intense research because of their many manufacturing applications.⁶¹ A generally accepted model describing the SiO_2 –water interface is given in Figure 19.5.⁶²

The SiO_2 lattice consists of siloxane ($\equiv \text{Si}-\text{O}-\text{Si} \equiv$) bonds. Upon immersion in water, these siloxane bonds can be hydrated to form silanol groups ($\equiv \text{Si}-\text{OH}$). Experimental evidence has shown the presence of two main types of silanol groups: (i) single silanols with one $-\text{OH}$ group attached to a Si lattice atom and (ii) geminal silanols consisting of two $-\text{OH}$ groups attached to one Si lattice atom. Furthermore, hydrogen bonds can be formed between adjacent water molecules and the polar hydroxyl groups. Because of the polarization of the $\text{Si}-\text{O}$ bond in the hydroxyl groups, the silanol groups can deprotonate^{59,63,64}:



or be protonated^{59,63}:



As a result of these reactions, the silica surface is generally covered with a negative surface charge, except at very low pH (typically $\ll 2-3$) where an excess of protons in the bulk results in a positive surface potential. This behavior is consistent with other mineral oxides that carry a positive surface charge below their point of zero charge.⁶⁵⁻⁶⁷

In spite of this seemingly simple representation of the silica–water interface, many phenomena are not yet clearly explained, such as the extraordinary stability of silica suspensions around the isoelectric point of the surface^{59,65,68} or the presence of a *gel* layer covering the silica surface.^{59,69,70} These

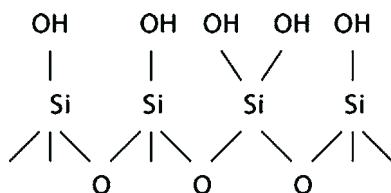


FIGURE 19.5 Schematic representation of a silica surface in contact with water.

peculiarities are often ascribed to the strong similarity that exists between the structure of water and of SiO_2 . Indeed, from a bulk point of view they both represent a close packing of large oxygen atoms into tetrahedral structures complemented by the smaller hydrogen and silicon atoms.^{59,63,71} Therefore, it is not surprising that there is a strong interaction between the water molecules and the silica surface, as evidenced by the SRR interactions described in Section 19.2.1.3.

It should also be mentioned that it is not straightforward to compare results obtained on various silica samples with each other. This is because differences in the pH and preparation methods of the silica suspensions influence their surface properties, such as the surface density of hydroxyl groups, the presence of contaminant atoms in the lattice, and other properties.⁵⁹

19.2.2.2 Interaction Between Metal Cations and a Silica Surface

19.2.2.2.1 Ion Exchange Model

The generally accepted mechanism for the adsorption of cations on a silica surface is through an ion exchange reaction at the surface hydroxyl groups.^{59,64,71} The equilibria governing this process can be written as:



where K_H and K_M are the equilibrium constants that describe the adsorption of H^+ and M^{n+} onto the surface, respectively. This model can generally be used irrespective of the form in which the ions are adsorbed on the surface (for example, in hydrated form or in a hydroxylated complex) and the surface composition (for example, whether the silanol groups are isolated or geminal). More complex models that take these effects into account have been proposed, but these must rely on certain assumptions of adjustable terms such as surface potential model and solvation free energy.^{65,71–73}

In this chapter, we will adopt the simple ion exchange model from Eqs. (19.10) and (19.11), since it has already been applied with success to study the adsorption of metal cations on silicon wafers covered with a chemical oxide by other researchers.^{74–77} In their calculations, the authors assume a finite concentration, σ_{max} , of available common adsorption sites onto which cations (M^{n+} and also H^+) can attach singly. For the simple case of only one cation (in addition to H^+), we can write the following surface concentration balance:

$$\sigma_{\text{max}} = \sigma_{\text{SiOM}} + \sigma_{\text{SiOH}} + \sigma_{\text{SiO}^-} \quad (19.12)$$

where σ_{SiOM} is the surface concentration of the metal on the wafer surface. Combining Eqs. (19.10)–(19.12), one obtains the following expression for the metal surface concentration:

$$\sigma_{\text{SiOM}} = \frac{K_M[M^+]}{1 + K_H[H^+] + K_M[M^+]} \sigma_{\text{max}} \quad (19.13)$$

which is equivalent to the Langmuir model for gas adsorption.²¹ This model can be easily extended to include the presence of more cations by summing the corresponding equilibria for each of the species.

19.2.2.2.2 Effect of pH

Equation (19.13) shows that the total surface concentration of metal ions depends on the pH of the solution. This is because in the model the protons and metal cations are considered to be competing for the available sites. Hence the amount of adsorbed metal cations depends on the relative amount of H^+ and M^+ ions. Assuming that evaporation does not depend on the pH of the liquid, we can distinguish between adsorption and evaporative deposition if we vary the pH of the solution.

19.2.2.2.3 Maximum Surface Concentration

The maximum surface concentration of species σ_{max} is also an important characteristic of the adsorption process, as it will give us an upper limit for the amount of metals on the surface due to adsorption. Theoretically, the density of silanol groups on an ideal fully hydroxylated β -cristobalite surface (one of the commonly encountered crystalline forms of silica in nature⁶⁴) is shown to be $4.55/\text{nm}^2$ or equivalently $4.55 \times 10^{14}/\text{cm}^2$.^{59,63} This number corresponds very well with experimental results on various types of silica that are reported in the range of $4\text{--}6/\text{nm}^2$.^{59,63,71,78} This value, however, represents a theoretical maximum. In practice, it has been observed that the amount of available adsorption sites is significantly less,^{59,71} indicating that not all surface hydroxyl groups act as possible exchange sites. Several reasons are given to explain this difference. First of all, cations in an aqueous environment are covered by a shell of water molecules (called the *hydration shell*) which increases their effective radius significantly.⁷¹ If the adsorption of the cation occurs in its hydrated state, the area occupied by such a hydrated ion can be larger than the average area occupied by an isolated silanol group (typically 0.2 nm^2). A second effect is that the effective dissociation constant of the remaining silanol groups decreases as more silanol groups become deprotonated.⁵⁹ As more metal ions are present on the silica surface, less sites become available for exchange. A similar conclusion has been reached from deuterium exchange experiments,⁷¹ which showed that the number of exchanged protons was much lower than the density of hydroxyls on the surface. This indicates that the surface density was limited by repulsion between electrostatic charges. Finally, it has been found that the degree of acidity of the silanol groups increases with crystallinity.⁶⁵ A detailed study of cation adsorption on hydrophilic silicon wafers predicted a maximum site density of approximately $3 \times 10^{12} \text{ atoms/cm}^2$.^{74–77,79} In the same papers,

however, values of up to 1.3×10^{13} K atoms/cm² after contact with a 10^{-5} M KCl solution were reported.⁷⁹ Other authors studying adsorption of metal ions from NH₄OH/H₂O₂ mixtures onto silicon wafers also report surface concentrations up to $1\text{--}3 \times 10^{13}$ atoms/cm².^{80,81} Here, we shall assume that in acidic solutions σ_{max} for a wet chemical oxide is on the order of 10^{13} atoms/cm².

19.2.3 Literature Models Describing Wafer Drying

19.2.3.1 Spin Drying

Although no systematic investigation of the spinning process with the aim of drying wafers has been reported as such, much information is available in studies of photoresist spin coating and in studies of the lubrication of rotating hard disk assemblies. Both fields have in common that they describe the behavior of a thin liquid layer on a rotating disk. Various models have been developed looking at the influence of surface roughness, surface topography, liquid viscoelastic properties, and liquid volatility on the process performance.

19.2.3.1.1 Model for a Non-Volatile Liquid

A simple analytical model has been derived by Emslie et al.⁸² This model gives the film thickness h of a non-volatile Newtonian liquid (with viscosity η and density ρ) on an infinite rotating disk as a function of spinning time:

$$h(t) = \frac{h_0}{\sqrt{1 + \frac{4\rho\omega^2 h_0^2 t}{3\eta}}} \quad (19.14)$$

where ω is the wafer rotation speed and h_0 is the initial film thickness which can be taken as a fitting parameter. Figure 19.6 gives the calculated thickness of a water film ($h_0 \cong 20$ μm , a typical value for an aqueous carry-over layer¹) for various rotation speeds.

An important finding of the work of Emslie et al. is that for a Newtonian liquid and axisymmetrical flow the film thickness on a spinning wafer does not depend on the radial position. Moreover, they have also shown that non-uniformities in the film will be leveled out by the spinning process.⁸² This is not the case for a non-Newtonian fluid which can result in non-uniform films.⁸³

19.2.3.1.2 Model for a Volatile Liquid

During the rotation process, however, a large flow of air is entrained over the wafer surface.^{84–86} Therefore, evaporation of the water film is likely to occur and should be taken into account as well. This is done in many of the models that describe the coating of silicon wafers with a photoresist film. Because such films contain a volatile solvent, evaporation is a crucial process parameter. However, the simultaneous solution of the necessary differential equations

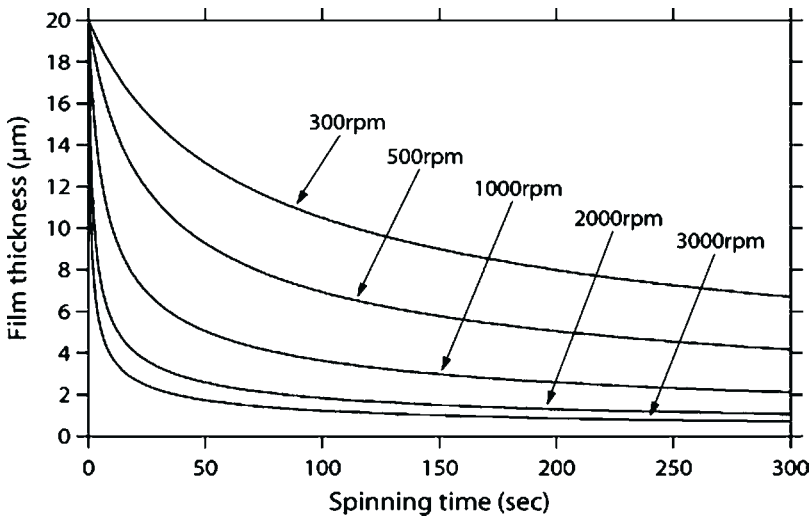


FIGURE 19.6 Calculated thickness of a water film with initial thickness $h_0 = 20 \mu\text{m}$ on a rotating wafer for various rotation speeds, according to Eq. (19.14).

for evaporation and convective liquid removal requires numerical simulations. Meyerhofer⁸⁷ concluded that two consecutive phases can be distinguished: (i) a first phase where spin-off (i.e. convective outflow) dominates and (ii) a second phase where evaporation takes over. Based on Meyerhofer's findings, Bornside et al.¹¹ showed that therefore the spinning process could be represented qualitatively as depicted in Figure 19.7. This figure plots the film thickness of a mixture containing a volatile solvent and a non-volatile solid (dispersed in the solvent) as a function of the spinning time. It is assumed that during spinning a uniform film thickness is maintained. At the end of the spin-off phase (at time $t = t_{\text{spin-off}}$) a film with thickness $h_{\text{spin-off}}$ remains. During the evaporation phase, the solvent evaporates and the film shrinks due to the reduction in volume until a film thickness h_{final} remains at a time $t = t_{\text{dry}}$.

Assuming that the volumes of solvent and solid are additive Meyerhofer showed that h_{final} did not depend on the initial thickness h_0 , as was the case with the model from Emslie et al.⁸² but could be written as:

$$h_{\text{final}} \propto \omega^{-2/3} \eta^{1/3} r_{\text{evap}}^{1/3} \quad (19.15)$$

with r_{evap} the volumetric evaporation rate per unit area (in m^3/s), which is assumed to be a constant. For a volatile liquid, the evaporation is limited by the mass transfer to the ambient gas, which is inversely proportional to the concentration boundary layer thickness in the gas phase $\delta_{\text{spin,c}}$:

$$r_{\text{evap}} \propto \frac{1}{\delta_{\text{spin,c}}} \quad (19.16)$$

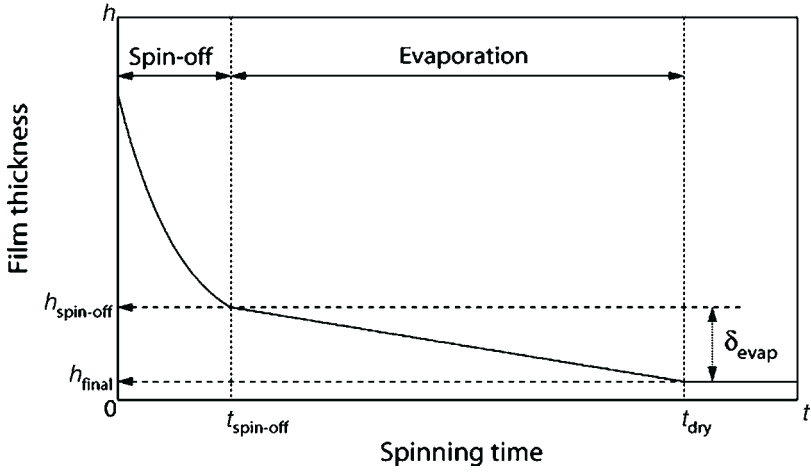


FIGURE 19.7 Qualitative representation of the film thickness of a mixture of solvent and a dispersed solid as a function of the spinning time according to Meyerhofer⁸⁷ and Bornside et al.¹¹

Because the Schmidt number $Sc = \nu_g/D_g$ (ν_g is the air kinematic viscosity and D_g is the liquid diffusivity in air) is nearly unity,¹² the concentration boundary layer thickness is comparable to the velocity boundary layer thickness $\delta_{\text{spin},v}$:

$$\delta_{\text{spin},c} \simeq \delta_{\text{spin},v} \quad (Sc \simeq 1) \quad (19.17)$$

with^{12,85}:

$$\delta_{\text{spin},v} \propto \sqrt{\frac{\nu_g}{\omega}} \quad (19.18)$$

Combining Eqs. (19.18) and (19.17) with (19.16) we can see that:

$$r_{\text{evap}} \propto \sqrt{\omega} \quad (19.19)$$

Substituting Eq. (19.19) into Eq. (19.15) we obtain:

$$h_{\text{final}} \propto \omega^{-1/2} \quad (19.20)$$

This relationship has indeed been confirmed by empirical data on spin coating of various photoresists^{88–91} and indicates that the evaporation of the solvent is limited by the entrainment in the gas phase. The volumetric gas flow rate entrained by the spinning wafer, Q_{spin} , is given by⁸⁵:

$$Q_{\text{spin}} = 0.885\pi R^2 \sqrt{\nu_g \omega} \quad (19.21)$$

with R the wafer radius. Comparing Eqs. (19.19) and (19.21), we can see that the evaporation rate is proportional to the entrained gas flow rate (since both are proportional to $\sqrt{\omega}$). This finding has also been used by Meyerhofer⁸⁷ in his model.

19.2.3.1.3 Film Uniformity

From the above derivations one can see that the position on the wafer surface does not play a role, that is, the models predict a uniform film thickness throughout the spinning process. However, it should be mentioned that these derivations are only valid when the flow conditions in the gas phase are in the laminar regime for small Reynolds numbers. For a rotating wafer the Reynolds number Re is defined as:

$$Re = \frac{R^2 \omega}{v_g} \quad (19.22)$$

In case the flow regime is not laminar, the evaporation rate is a function of the position on the wafer surface and the resulting film profile will be non-uniform.^{12,85,88,92} According to an analysis by Schlichting⁸⁵ the transition from laminar to turbulent flow occurs at a critical Reynolds number Re of 3×10^5 . This corresponds to a rotation speed of 4297 rpm for a 200 mm wafer (using $v_g = 15 \times 10^{-6} \text{ m}^2/\text{second}$,⁹³ which corresponds to air at 20 °C). This exceeds the maximum rotation speed investigated in the experiments reported in this chapter. Bornside et al.¹², however, suggest that a transient range can occur during which spiral vortices (referred to as *Eckman spirals*) are formed. They claim that such a transient range can start at $Re = 1 \times 10^5$ (corresponding to 1432 rpm for a 200 mm wafer) or in some cases for Reynolds numbers as low as 6×10^4 (corresponding to 859 rpm for a 200 mm wafer). These conditions are within the experimental range used in the experiments reported here. Additionally, it has been shown by Bornside et al.¹² that the flow field above the wafer surface (and hence the evaporation rate) is influenced by the catch cup surrounding the rotating wafer and the exhaust suction employed. Finally, it has been shown that the acceleration of the spinner is also an important parameter. In their analysis of the uniformity of photoresists films, Flack et al.⁹⁴ reported that when the acceleration is not rapid enough, flow instabilities can occur, resulting in an uneven film profile. The non-uniformity of a coated film is, therefore, not necessarily an indication of a turbulent gas flow, but can also be ascribed to a combination of the spinner characteristics and the liquid properties.

19.2.3.2 Vertical Marangoni-Based Wafer Drying

As depicted in Figure 19.2, vertical Marangoni-based drying takes place when a tensioactive species is delivered from the gas phase into the liquid meniscus region. In the literature, however, most of the studies investigating surface tension gradients consider a system where the tensioactive component is initially present in the liquid phase. For such a process, the withdrawn film thickness is predicted to increase under the influence of the Marangoni effect.^{95–97} This phenomenon is in contrast to Marangoni-based drying, where the film thickness is reduced.⁹⁸ Very limited studies of Marangoni-based drying are available, mainly due to the complexity in simultaneously solving the equations of mass

transfer, diffusion, and convection. We shall limit our discussion to a published model for Marangoni-based drying⁹⁸ that is in qualitative agreement with the trends that have been observed empirically by Leenaars et al.,³² namely that the Marangoni effect results in a reduced film thickness. This model developed by Thess and Boos⁹⁸ allows one to calculate the thickness of the liquid layer entrained with a plate moving vertically at a speed U under the assumption that the liquid surface tension can be written as:

$$\gamma(x) = \gamma_0 - \bar{\tau}x \quad (19.23)$$

Here γ_0 is the surface tension of the liquid in the bath, $\bar{\tau}$ is the mean surface tension gradient ($= -\partial\gamma/\partial x$), and x is the height above the liquid level in the bath. In this model, the surface tension gradient is assumed to originate from a concentration gradient of a tensioactive compound $\partial C/\partial x$ with $C(x)$ the local concentration of tensioactive species. To simplify the mathematics, $\bar{\tau}$ is assumed to be constant over the meniscus region. The result of this calculation is an expression for the film thickness $h_{\text{Marangoni}}$ entrained with a solid surface:

$$h_{\text{Marangoni}} = H(Ma) \sqrt{\frac{\gamma}{\rho g}} Ca^{2/3} (Ca \ll 1) \quad (19.24)$$

where the dimensionless capillary number Ca (defined as $Ca = \eta/U\gamma$) expresses the relative importance of viscous forces over surface tension forces. It should be mentioned that Eq. (19.24) is very similar to the solution obtained by Landau and Levich⁹⁹ for the liquid thickness entrained with a solid substrate that is withdrawn from a liquid bath, in the absence of a surface tension gradient. The solution they obtain is:

$$h_{\text{LL}} = 0.9458 \sqrt{\frac{\gamma}{\rho g}} Ca^{2/3} (Ca \ll 1) \quad (19.25)$$

The difference between Eqs. (19.24) and (19.25) is the numerical constant in Eq. (19.25) which is replaced with the dimensionless thickness function $H(Ma)$. The dimensionless Marangoni number Ma is defined as

$$Ma = \frac{\bar{\tau}}{\gamma^{1/6} g^{1/2} \rho^{1/2} U^{1/3} \eta^{1/3}} \quad (19.26)$$

and expresses the relative importance of surface tension gradient effects over viscous effects. It should be noted, however, that in the literature other definitions of the Marangoni number also exist. This makes the comparison difficult between the reports from different authors.¹⁷ By taking the ratio of Eqs. (19.25) and (19.24), we can see that $0.9458/H(Ma)$ represents the reduction in thickness that is obtained when a surface tension gradient is present, as compared to the withdrawal process without surface tension gradients. The value of H as a function of Ma is represented graphically in Figure 19.8 by the black markers that correspond to the values listed in Table 19.2 from Ref. 98.

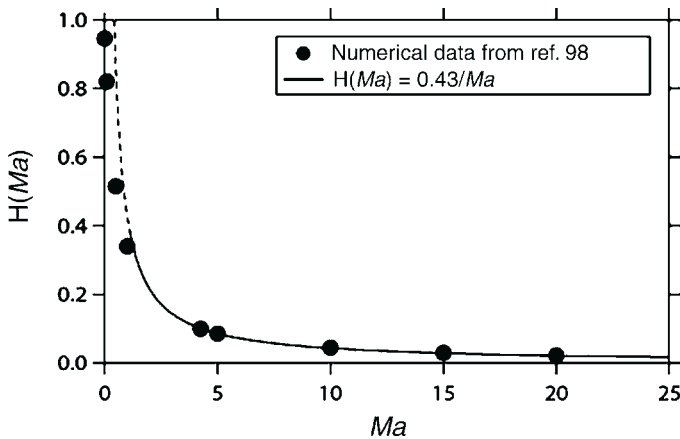


FIGURE 19.8 Dimensionless thickness function $H(Ma)$ from Eq. (19.24) as a function of the dimensionless Marangoni number (defined in Eq. (19.26)). The markers represent data reported by Thess and Boos.⁹⁸ The solid curve represents Eq. (19.28).

Figure 19.8 shows that $H(Ma)$ is a monotonously decreasing function. For small values of Ma , the Marangoni effect becomes negligible, and Eq. (19.24) reduces to Eq. (19.25):

$$H(Ma) = 0.9458(Ma \ll 1) \quad (19.27)$$

For large values of Ma , the following asymptotic expression is given by Thess and Boos⁹⁸:

$$H(Ma) \approx \frac{0.43}{Ma}(Ma \gg 1) \quad (19.28)$$

Equation (19.28) is also plotted in Figure 19.8 as a solid line. It is clear that the asymptotic equation provides a good description for values of Ma larger than 2, or alternatively, for those conditions where the resulting film thickness is less than half of the thickness without surface tension gradient. Furthermore, it can be seen that upon substitution of Eq. (19.28) into Eq. (19.24) we obtain:

$$h_{\text{Marangoni}} \approx 0.43 \frac{U\eta}{\bar{\tau}}(Ma \gg 1) \quad (19.29)$$

This equation shows that for large Marangoni numbers the entrained thickness no longer depends on gravity and surface tension, but only on viscosity, withdrawal speed, and surface tension gradient. Because there is a direct correlation with the entrained film thickness, the mean surface tension gradient $\bar{\tau}$ theoretically can be used as a performance indicator for a surface tension gradient-based drying technique.

19.2.3.3 Applicability of the Model for Vertical Drying to a Rotating Wafer System

Because of the complexity of the calculations involved, a physical model to describe Marangoni-based drying on a rotating wafer is not derived here. There is, however, considerable similarity between Marangoni-based drying on a rotating wafer and on a wafer that is being vertically lifted, allowing us to use the same model under certain conditions. To understand this similarity, consider an observer rotating at the same speed as the wafer and moving from the wafer center to the wafer edge at the same speed as the nozzles (see also Figure 19.11). To that observer, the meniscus region will appear static, while the wafer is moving with a speed U (opposite to, but with the same magnitude as the nozzle speed). Because the coordinate system of the observer is non-inertial, we must add centrifugal $\left(\vec{\omega} \times \left(\vec{\omega} \times \vec{r}\right)\right)$ and Coriolis accelerations $\left(2\vec{\omega} \times \vec{v}\right)$ to the fluid, where $\vec{\omega}$, $\vec{r}$, and $\vec{v}$ are the vectors describing the wafer rotation, nozzle position (with reference to the rotating wafer), and nozzle movement (with reference to the rotating wafer), respectively. In normal operating conditions, the nozzle speed is significantly lower than the circumferential speed at each point (typical values are a rotation speed on the order of a few 100 rpm and a nozzle speed on the order of a few mm/s). Under these conditions, the nozzles are perceived by the observer as two concentric circular slits with a relative movement $\vec{v}$ that is directed away from the axis of rotation (if the nozzles cannot be considered as circular slits, but instead as point sources, their relative movement would be an inwardly directed spiral). Under these typical operating conditions, the Coriolis effect is less than 1% of the centrifugal acceleration. Additionally, there is also the earth's gravitational acceleration, which is acting perpendicularly to the direction of the nozzle movement and is flattening out the liquid film. However, for the typical rotating speeds used, the centrifugal acceleration is significantly higher than the gravitational acceleration, about 10 times higher at the edge of a 200 mm wafer rotating at 300 rpm. Therefore, in a first approximation we can neglect the effect of Coriolis and gravitational acceleration. For this system we can use, therefore, the same mathematical solutions as for the vertical Marangoni-based drying technique. It must be mentioned, however, that the boundary conditions used in the mathematical description of this system require the liquid bulk to be static and infinitely large. For the simplified system of a rotating wafer, this means that turbulence vortices in the liquid arising from the impinging liquid delivered through the second nozzle are neglected. Furthermore, it must be assumed that the thickness of the liquid pool at the point of dispensation is very large compared to the meniscus dimensions.

Assuming the simplifications do not significantly alter the predictions of the mathematical model, we can expect the same qualitative behavior as described by Eqs. (19.24) through (19.29) for a rotating wafer. Interestingly enough, these equations predict that if the Marangoni effect is sufficiently strong ($Ma \gg 1$), the

film thickness left on the wafer after rotational Marangoni-based drying depends only on the nozzle speed, liquid viscosity, and the surface tension gradient induced by the flows delivered through both nozzles. It is important to realize that according to Eq. (19.26), where the gravity constant g must be replaced with $\omega^2 r$, the Marangoni number decreases with the square root of the radial position. Eq. (19.24) shows, however, the same dependence on r in the denominator of the fraction on the right-hand side of the equation. Consequently, the model predicts that when the Marangoni number is high the film thickness for a rotational Marangoni-based drying process is independent of the radial position on the wafer.

19.2.3.4 Limitations of the Model for Marangoni-Based Drying

19.2.3.4.1 Surface Tension Gradient

One of the limitations of the model⁹⁸ is the fact that the mean surface tension gradient $\bar{\tau}$ is being imposed as a boundary condition instead of being correlated with the mass flow of tensioactive vapor to the surface. Because $H(Ma)$ is a monotonously decreasing function, there is a one-to-one correlation between the final film thickness and the mean surface tension gradient $\bar{\tau}$. This means that if one has a measurement of the final film thickness, one can use the model to calculate the mean surface tension gradient for a given Marangoni-based drying process. The result of these calculations will be given in Tables 19.1 and 19.4 for data obtained from the literature and from experimental data in this chapter, respectively. When no experimental data are available, however, it is impossible to use the model in a predictive manner.

19.2.3.4.2 Surface Forces

A second limitation of the model lies in the assumption that the fluid properties are assumed to be constant up to the contact with the substrate. For large Marangoni numbers, that is, the normal process window targeted by a Marangoni drying step, the actual film thickness can be smaller than 1 μm . For these films, as discussed in Section 19.2.1.1, the intermolecular forces can dominate the behavior of the liquid film. An example has been given in Section 19.2.1.3 on the extraordinary behavior of water films on a silica surface.

19.2.4 Salt Tracer Tests

As discussed in Section 19.1.2, the main body of this work is focused on the use of salt tracer tests, whereby information on the drying step is derived from the salt residues that are left on the wafer surface after the wafer has been in contact with a salt solution of known concentration and dried. During such a test, two different deposition mechanisms (adsorption and evaporative deposition) take place. For a correct interpretation of the raw data it is, therefore, necessary to find a way to distinguish between these two mechanisms. This approach will be

TABLE 19.1 Summary of Literature Data Describing the Drying Performance of a Vertical Marangoni Drying Process. δ_{evap} : Experimentally Determined Entrained Thickness Taken from Ref. 98

Experimental Condition	U (mm/s)	δ_{evap} (μm)	h_{LL} (mm)	$\bar{\tau}$ (N/m ²)	Ma (—)
Directed vapor stream	0.7	0.014	1.1	19	33.5
Directed vapor stream	1.5	0.014	1.8	40	54.7
Semiquiescent atmosphere	0.7	0.11	1.1	2.4	4.2
Semiquiescent atmosphere	1.5	0.16	1.8	3.6	4.9

h_{LL} : Entrained Thickness for the Same Withdrawal Speed U , but Without Marangoni Effect Calculated from Equation (19.25). $\bar{\tau}$: Mean Surface Tension Gradient Calculated from Equation (19.29) Taken from Ref. 98. Ma : Marangoni Number Calculated from Equation (19.24).

explained first. Next, we shall also present some literature data that have been obtained using a comparable approach to investigate the drying process.

19.2.4.1 Interpretation of Salt Tracer Tests

The essence of salt tracer tests consists in relating the surface concentration of the salt tracer species on the wafer after drying to its bulk concentration in the liquid that is brought into contact with the wafer. This relation is different for the two types of deposition mechanisms proposed in Section 19.1.2. For the adsorption process (see Eq. (19.13)), a Langmuir type of behavior has been proposed. The main characteristics of this model as it is used in this work are: (1) there is a pH dependence of the surface concentration, due to the competition between the H^+ and metal ions and (2) there is a maximum surface coverage of the order of 10^{13} atoms/cm². For the evaporation process (see Eq. (19.4)), the surface concentration increases linearly with the bulk concentration, assuming that δ_{evap} remains constant. When both contributions are plotted schematically on the same graph we obtain Figure 19.9 (a and b denote linear and logarithmic axes, respectively).

Figure 19.9 shows that at high salt concentration, the surface concentration due to evaporative deposition is dominant. In this range (evidenced by the fact that $\sigma \gg 10^{13}$ atoms/cm²), the relation between surface and bulk concentrations of metal ions is expected to be linear and δ_{evap} is obtained by taking the ratio of both values (Eq. (19.5)). Figure 19.9 also shows that at low concentrations adsorption is the dominant mechanism. Furthermore, in this concentration range, the surface concentration of metal ions is a function of the pH of the

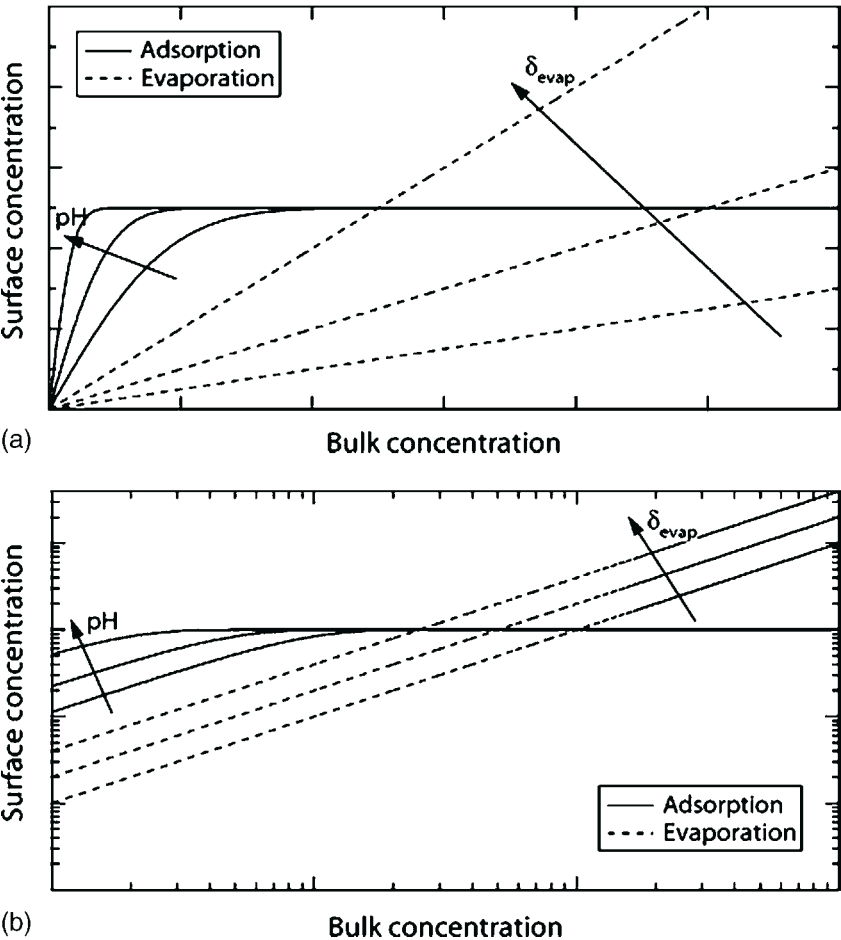


FIGURE 19.9 Schematic illustration of the use of metal salt tracer tests. Solid lines represent adsorption of metal ions according to the Langmuir adsorption isotherm (Eq. (19.13)), dashed lines represent deposition of metal ions as a result of liquid evaporation (Eq. (19.4)), characterized by an equivalent evaporated thickness δ_{evap} on (a) a linear scale and (b) on a logarithmic scale. The plateau in the adsorption isotherms corresponds to a surface concentration on the order of 10^{13} atoms/cm².

solution (when it is assumed that the pH does not influence the drying process, an observed pH dependence can also be used as an indirect proof that adsorption is responsible for the deposition of metal cations on the surface).

19.2.4.2 Available Literature Data

19.2.4.2.1 Spin Drying

The procedure for dispensing metal salt solutions on a hydrophilic rotating wafer has already been used for quite some time in the semiconductor industry

as a method to prepare silicon wafers with a known amount of metallic impurities,^{100–102} and also as a method to study adsorption on silicon wafers, as already discussed in Section 19.2.2.^{74–77,79} However, until now no systematic study of the relative contributions of adsorption and evaporation has been reported.

19.2.4.2.2 Marangoni-Based Drying

The use of a metal salt solution to evaluate the performance of a Marangoni-based drying technique has been proposed by Marra and Huethorst.³³ However, they did not take adsorption into account in their tests. A summary of their results is given in Table 19.1 for two types of gas flow (a directed flow and a semiquiescent ambient) and for withdrawal speeds U of 0.7 and 1.5 mm/s.³³ These tests were performed using a 0.01 M CoCl_2 solution. In Table 19.1, δ_{evap} is the experimentally determined thickness reported by Marra and Huethorst³³ calculated from Eq. (19.5). We use the notation δ_{evap} instead of $h_{\text{Marangoni}}$ assuming that all entrained liquid evaporates. h_{LL} represents the solution to Eq. (19.25) without a surface tension gradient present.

The same data have been used later by Thess and Boos⁹⁸ to calculate the corresponding mean surface tension gradient $\bar{\tau}$, which is included in the table. In the final column we have also added the corresponding Marangoni numbers based on Eq. (19.26).

19.3 EXPERIMENTAL DETAILS

19.3.1 Setups for Wafer Drying

In this section, an overview of the various drying platforms used in this work is given. They can be divided into spin drying platforms and surface tension gradient based (or Marangoni based) drying platforms. Marangoni-based platforms used in these tests include vertical as well as horizontal (rotational) setups.

19.3.1.1 Spin Drying

The two spin drying platforms (capable of drying 200 mm wafers) will be denoted as “manual” and “semi-automated” referring to the method of loading the wafers and dispensing the liquids during rotation. In both the cases the actual spin dry process is fully automated.

19.3.1.1.1 Manual Spinner

The manual tool used in this work was a model WS-400 spinner capable of handling wafers up to 400 mm in diameter. The wafer is loaded manually on a chuck—hence we refer to this setup as the *manual* system—and held in place using vacuum. A plastic ring is positioned around the wafer to trap outwardly flying droplets.

19.3.1.1.2 Semi-Automated Spinner

The second drying tool used in this work was the platform of a Goldfinger™ cleaning system. In this system, the wafers are manually loaded onto a chuck containing three equally spaced carrier pins. Before starting the process, the chuck is lowered into a process bowl that serves as a splash guard to trap outwardly flying droplets. A special mechanism with counterweights automatically holds the wafer in place when rotation is started. Because this system allows automatic dispensation of liquids during the process as well as controlling the movement of the dispensation arm, we refer to it as a semi-automated spinner. Visual observations also suggest that the motor of this system is capable of delivering a higher acceleration than the one from the manual tool.

19.3.1.2 Vertical Marangoni-Based Drying

19.3.1.2.1 Batch Marangoni-Based Dryer

For simultaneous drying of multiple wafers, a 200 mm wafer commercial *Marangoni Dryer*™ was used. In this system, a Teflon® carrier holding the wafers is lowered in a tank filled with ultrapure water (UPW). Next, a hood is positioned over the tank and saturated with a mixture of IPA in nitrogen. The wafers are then slowly pushed out of the liquid by means of a special knife with minimal contact. It should be noted that this setup was used only in the preparation (preclean) of the wafers prior to testing (as described in [Section 19.3.3.1](#)). Due to contamination restrictions, the salt residue tests were not performed on this platform.

19.3.1.2.2 Single Wafer Marangoni-Based Dryer

For single wafer Marangoni-based drying, a prototype was designed and fabricated. A schematic drawing of this setup is given in [Figure 19.10](#).

The setup consists of a tank with inner dimensions of approximately 30 cm wide by 30 cm high and 5 cm deep. The back plate is made of polyvinyl chloride (PVC) and the front plate of transparent poly (methyl methacrylate) (PMMA) to allow visual inspection of the drying process. The PVC top plate contains a narrow (≈ 10 mm) slit through which the wafers leave the bath. During drying, a vertical liquid flow (≈ 11 per minute) is activated to deliver fresh liquid to the meniscus region. The excess liquid flows over a thin rim in the top plate and is pumped to a drain. The wafer carrier is made from polyether ether ketone (PEEK) and is specially designed to have a minimal number of contact points (see [Figure 19.10](#)). These contact points are all located at the bottom half of the wafer, at the upper half no contact is made with the wafer in order not to disturb the flow pattern when it is lifted from the tank. A few cm above the overflow rim two PVC nozzles are positioned (one on each side of the wafer) directed at the position where the wafer leaves the liquid. These nozzles are slit-shaped with a width of 0.5 mm and a length of 20 cm. A N_2 /IPA vapor is delivered into both

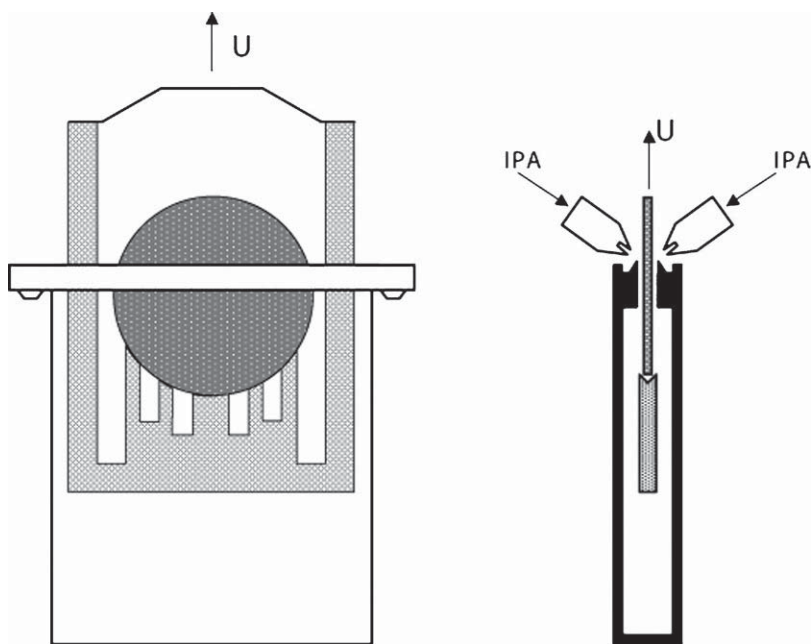


FIGURE 19.10 Schematic drawing of the set-up used to perform vertical Marangoni based drying tests on a single 200 mm wafer.

nozzles. This is done by blowing ultrapure N_2 gas through a bubbler system filled with liquid IPA at room temperature. The mixture is then equally divided between the two nozzles. The flow of nitrogen is adjusted by a mass flow controller (Brooks Instrument Model 5850 s) with a maximum flow of 20 Standard Liters per Minute (slm). Withdrawal of the wafers is done by means of a gear wheel mechanism connected to a DC motor. The withdrawal speeds attainable in this way ranges from 1.4 to 35 mm/s.

19.3.1.3 Marangoni-Based Drying of Horizontally Rotating Wafer

Marangoni-based drying of a rotating wafer (licensed under the name *Rotagoni*TM) is performed on the two spin dryer platforms (see Section 19.3.1.1) by equipping them with a system of translating nozzles.^{9,37–40} A schematic description is given in Figure 19.11.

The tensioactive vapor is delivered to the wafer surface through a small nozzle (inner diameter 1 mm) from a distance of about 10 mm from the wafer surface, aimed perpendicularly at the wafer surface. This vapor is formed in the same way as on the vertical IPA-assisted drying technique. During drying, water is dispensed through the second nozzle (with an inner diameter 1 mm) that is positioned at the same height above the wafer, but a few mm further from the center. This nozzle is also slightly tilted in the same direction as the wafer rotation

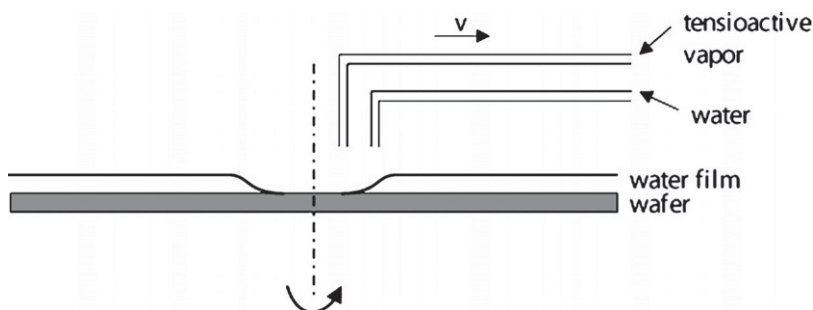


FIGURE 19.11 Schematic drawing of Marangoni-based drying on a horizontally rotating wafer.

(i.e., tangentially), to avoid splashing of the liquid when it hits the wafer surface. The water flow rate is on the order of 100 ml/min. Drying of the wafer is accomplished by positioning both nozzles over the center of the wafer and subsequently moving both nozzles at the same speed toward the edge of the wafer. The liquid flow is stopped a few mm before the nozzle reaches the edge of the wafer, to avoid splashing when the water flow hits the wafer bevel.

19.3.1.3.1 Manual Tool

For the implementation of Marangoni-based drying on the manual spinner the nozzles are mounted onto a moving piston, powered by compressed air. The piston speed can be adjusted by a flow valve that regulates the oil damping inside the piston. Pneumatic valves in the feed lines of the water and nitrogen (to the bubbler system) are activated manually as well.

19.3.1.3.2 Semi-Automated Tool

For the implementation on the semi-automated tool, two nozzles are mounted onto a moving arm powered by an electrical gear-wheel mechanism. Using a micro-controller system, the entire drying process (including rotation, liquid and vapor delivery and translation) can be programmed and executed automatically after loading the wafer.

19.3.2 Analytical Techniques

19.3.2.1 Metal Surface Concentration by TXRF

19.3.2.1.1 Technique

The surface concentration of metallic species on a wafer is measured using Total Reflection X-Ray Fluorescence Spectroscopy (TXRF).^{103–105} In this technique, the wafer surface is irradiated by a primary monochromatic collimated X-ray beam generated by bombardment of electrons from a filament onto a metal anode such as W or Mo. The incident angle (between the beam and the wafer surface) is

TABLE 19.2 Details of the Three TXRF Systems Used in this Work

System Number	1	2	3
System	Atomika 8010	Atomika 8300(a)	Atomika 8300(b)
Primary X-ray beam	Mo-K α	Mo-K α	W-L β
Primary X-ray energy	17.44 keV	17.44 keV	9.67 keV
X-ray tube settings	50 kV, 40 mA	50 kV, 55 mA	50 kV, 55 mA

1.3 mrad which is smaller than the critical angle for total reflection. As a result the penetration depth of the primary X-rays is on the order of a few nm only (as compared to 10 μm for conventional XRF systems¹⁰³). This makes the TXRF technique very sensitive to elements located at the surface of the specimen. The primary X-rays excite the atoms in the sample to fluorescence. These secondary X-rays are detected by a Si(Li) energy dispersive detector. Because the energy of these secondary X-rays is specific for each element various elements can be measured simultaneously. From the detected X-ray intensity of each peak a quantitative result is obtained, using the measured intensity of a Ni standard calibration wafer (with a microdroplet of Ni) in combination with a built-in database of relative sensitivity factors. Because of availability reasons, three different TXRF systems were used throughout this study. Experimental details of these systems are summarized in [Table 19.2](#).

19.3.2.1.2 Sources of Error

To allow comparison between the results measured on different systems, it is important to consider the various sources of error in the measurements. A first error is related to the manufacturer and the exact model of the TXRF system and its corresponding calibration procedure (micro-droplets or spin coated samples^{102,105–107}). A detailed 2-year round robin test comparing various TXRF tools revealed that this error can be on the order of 60%.¹⁰⁸ A comparison of KCl coated wafers measured on systems 2 and 3 showed in our case deviations between the two tools from 20% to 50%. A second error is related to the fact that the measurement is influenced by the way in which the species of interest is present on the surface: as a smooth film or as particulate matter. To minimize the error associated with this geometric distribution, the TXRF measurements are performed at the so-called *iso-angle*, where ideal film-like and ideal particle-like samples behave in a similar manner.^{103,108} In reality, however, samples exhibit a certain micro-crystallinity that depends on the preparation method (adsorption from solution, spin coating, droplet drying, etc.^{101,102,107}) and on the total concentration. From angle scans it has been shown that samples possessing such micro-crystallinity can result in quantification errors up to

40%.¹⁰⁸ Finally, there are also random errors that occur when the same wafer is measured several times on the same tool. In this case, however, it is difficult to distinguish between errors related to sample stability and errors related to drift of the tool. For example, in the case of the KCl tests mentioned above, there was a time delay of 2 weeks between the two measurements which may partly be caused by these random errors. Experiments on Ni coated wafers showed an increase of the measured surface concentration by 23% in about 2 weeks, when measured on the same wafer and location. This value is in agreement with reported data that also show differences on the order of 20%¹⁰⁸ for measurements performed at various times.

For a correct interpretation of the experimental results of this work it is, therefore, important to distinguish between the accuracy and the precision of the measurements.¹⁰⁹ Accuracy refers to the deviation between the measured value and the actual value¹⁰⁹ (as would be obtained from an absolute technique, such as Rutherford Backscattering^{107,108}). Precision is a measure of the expected deviation between the measured value and the mean value, obtained in exactly the same way.¹⁰⁹ The round robin results showed that the accuracy of the TXRF tools used in this work (given in Table 19.2) is of the order of 40%. This means that the reported values can deviate 40% from the “real” value. From experimental results obtained on the tools, we can conclude that for identical conditions the precision is of the order of 10–20%. It must be mentioned that “under identical conditions,” we mean that the measurements are performed on the same tool, using the same measurement settings. Unless specified otherwise in the text, the results given in each graph are measured under identical conditions and a standard deviation of 10–20% can be assumed. When results reported on different graphs are compared, an accuracy of approximately 40% should be assumed. In case multiple measurements are performed on the same wafer, the standard deviation calculated from the measurements is indicated on the graph. Because the experimental conditions are chosen over a wide range, accessible by straight TXRF measurements ($\approx 10^{11}$ atoms/cm² up to several 10^{15} atoms/cm², depending on the element, primary X-rays and the tool), the estimated precision and accuracy are sufficient for our studies.

19.3.2.2 Surface Tension by Wilhelmy Plate Method

The surface tension of a liquid (γ) is determined using the Wilhelmy plate method.²¹ In this technique, a thin plate suspended on a sensitive balance is lowered into the liquid of interest. The total force on the balance can be expressed as:

$$F_{\text{tot}} = F_{\text{plate}} + \gamma L_{\text{perim}} + F_{\text{buoyancy}} \quad (19.30)$$

where F_{plate} is the weight of the thin plate, F_{buoyancy} is the buoyancy force, and γL_{perim} is the force exerted by surface tension on the immersed solid (with L_{perim} the perimeter of the plate). Note that it is implicitly assumed that the contact angle is zero. In order to obtain such a low contact angle, the measurements are

performed using a Pt plate that is thoroughly cleaned with chromic acid. For aqueous IPA mixtures this results in an extremely low contact angle. When the plate is immersed in the liquid, the total force decreases due to buoyancy forces that push the immersed volume upwards. This effect is proportional to the immersed volume. By extrapolating the curve of the force vs. immersion depth toward zero immersion, one can calculate the surface tension of the liquid. In this work, a model K12 Wilhelmy plate setup was used. The corresponding software automatically calculates the surface tension by regression of the measurements points.

19.3.3 Characterization of Materials and Products

19.3.3.1 Wafers

The wafers used for testing were p-type 200 mm $\langle 100 \rangle$ silicon wafers from Wacker Siltronic with a thickness of $725 \pm 15 \mu\text{m}$ and resistivity of 1–100 $\Omega \text{ cm}$. All wafers receive an IMEC clean³⁶ a few hours prior to being used. This procedure was executed in an automated wet bench and consisted of several consecutive steps: (i) $\text{H}_2\text{SO}_4/\text{O}_3$ (to remove organic contamination), (ii) hot quick dump rinse, (iii) HF/HCl (to remove particulate and metallic contamination), (iv) dilute HCl/O_3 (10 ppm) (to remove metallic contamination), and (v) Marangoni drying. Following this IMEC cleaning procedure, a thin silicon oxide layer (thickness between 0.5 and 1 nm) was grown on the silicon surface. The surfaces of these wafers are also commonly referred to as *ozone-last* surfaces, referring to the final contact with ozone present in the final rinsing step.¹¹⁰

19.3.3.2 Liquid Chemicals

The water used in all experiments was Ultrapure Water (UPW) with a conductivity of 0.055 $\mu\text{S/m}$ (or resistivity of 18.2 $\text{M}\Omega \text{ cm}$). The pH of the solutions was adjusted with 37% hydrochloric acid (HCl) with a specification of less than 10 ppb for most common metals.

19.3.3.3 Metal Salts

Metal salts used in the tests were KCl and $\text{Ni}(\text{NO}_3)_2$. The KCl solutions were prepared from 99.99% purity powdered salt. The $\text{Ni}(\text{NO}_3)_2$ solutions were prepared from a 1000 ± 2 wppm liquid Ni standard Atomic Absorption Spectroscopy (AAS) solution containing 0.5 M HNO_3 .

19.3.4 Procedure for the Salt Tracer Tests

19.3.4.1 Spin Drying

First, the solution is dispensed onto the wafer. For tests on the manual spinner (see Section 19.3.1.1.1), this is done by pipetting manually a sufficient amount of liquid to cover the entire wafer surface (on the order of 30 ml). For tests on

the semi-automated tool (see [Section 19.3.1.1.2](#)), this is done by means of the dispensation nozzle connected to the tool. Because the initial film profile does not affect the final evaporated thickness (as shown by Eq. (19.15)) we expect that the dispensing procedure does not play a role. This has been confirmed experimentally.¹⁰² Next, the wafers are spun around their axis until the surface is visually dry, which can be seen from the interference fringes that occur as the liquid is evaporating and the film is thinning. It must be mentioned that for the tests using a low salt concentration (in the ppb range), a waiting period of 60 s is introduced before starting the spinning process. This allows sufficient time for ions to diffuse to the surface.^{74,79} Finally, the surface concentration is determined by TXRF at various distances from the wafer center. This value is converted to an equivalent evaporated thickness δ_{evap} by means of Eq. (19.5).

19.3.4.2 Vertical Marangoni-Based Drying

The vertical Marangoni drying salt tests were performed in the setup described in [Section 19.3.1.2.2](#), which is depicted in [Figure 19.10](#). In these tests, the tank is filled with a metal salt solution. The wafer is loaded in the single wafer carrier and immersed in the solution for 30 s. At the start of the drying process, a gentle overflow (on the order of 1 l per minute) is started and the N_2/IPA flow is turned on. Next, the wafer is slowly lifted from the tank using a DC motor. Finally, the wafer is measured by TXRF on various equidistant positions along two perpendicular lines, parallel and perpendicular to the direction of withdrawal.

19.3.4.3 Marangoni-Based Drying on a Rotating Wafer

The salt tests on a Marangoni drying process on a rotating wafer were performed on the setups described in [Section 19.3.1.3](#) from which a schematic drawing of which is given in [Figure 19.11](#). The wafer is first pre-wetted by dispensing the salt solution on the wafer surface. Next, the drying process is started. It is important to note that during this drying process the same salt solution is delivered through the second nozzle (which normally serves to dispense UPW to the wafer). This is done to avoid dilution effects of the initial metal salt layer by the liquid flow from the second nozzle of the system.

19.4 RESULTS OF SALT TRACER TESTS

19.4.1 Spin Drying

19.4.1.1 Spin-Off vs. Evaporation

First, a visual investigation of the spin drying process was performed. It is seen that in the first few seconds, a large amount of liquid is spun off the wafer and thrown against the splash shield surrounding the rotating wafer. Subsequently, circular interference fringes are observed indicating the presence of a thin,

TABLE 19.3 Visual Drying Time of a Water Film on a Rotating 200 mm Wafer for Various Rotation Speeds

ω (rpm)	Drying Time
300	74
500	45
1000	26
1500	18
2000	15

axisymmetrical water film. As spinning continues, these fringes move outwards until, after a certain time, no more discolorations are observed. At this point, the film is at least significantly thinner than half of the wavelength of visible light. The elapsed time since the start of the spinning process, which we shall denote by the “visual drying time”, is listed in Table 19.3 for various rotation speeds.

Based on the calculated results plotted in Figure 19.6, we can see that for the typical drying times listed in Table 19.3, the model predicts that the films are still on the order of several micrometers thick. This is not in agreement with our visual observations, which suggest that liquid evaporation during drying is significant. To confirm this statement, metal salt tracer tests were performed on the semi-automatic tool using a 1000 ppm KCl solution. After dispensing the salt solution, the wafers are spun at 500 rpm. At various times, the spinning process is interrupted. In case the wafer still contains a liquid film, this film is left to evaporate in the ambient air till dry. Afterwards, the K surface concentration is measured by TXRF and converted to the corresponding evaporated thickness δ_{evap} using Equation (19.5). The results (based on an average of 13 equidistant points measured on two perpendicular lines through the wafer center) are given in Figure 19.12. The two y-axes on this figure are related through Eq. (19.5) (surface concentration on the right axis and the calculated evaporated film thickness on the left axis). The vertical error bars indicate the standard deviation on the 13 measurement points, while the horizontal error bars (≈ 3 s) represent the uncertainty in the acceleration and deceleration of the spinner. On the right axis, one can see that for all data points the total surface concentration is significantly higher than 10^{13} atoms/cm². This suggests that our assumption to neglect adsorption (using Eq. (19.5) to calculate the left axis) is justified.

The data show that during the first 10–15 s of spinning δ_{evap} decreases very fast, which is a result of the convective removal of the salt solution from the wafer surface. This corresponds to the spin-off stage, as indicated in Figure 19.7 (hence at 500 rpm $t_{\text{spin-off}} \approx 15$ s). From Table 19.3, we can see that the visual drying time is 45 s. Assuming that this value is close to the time

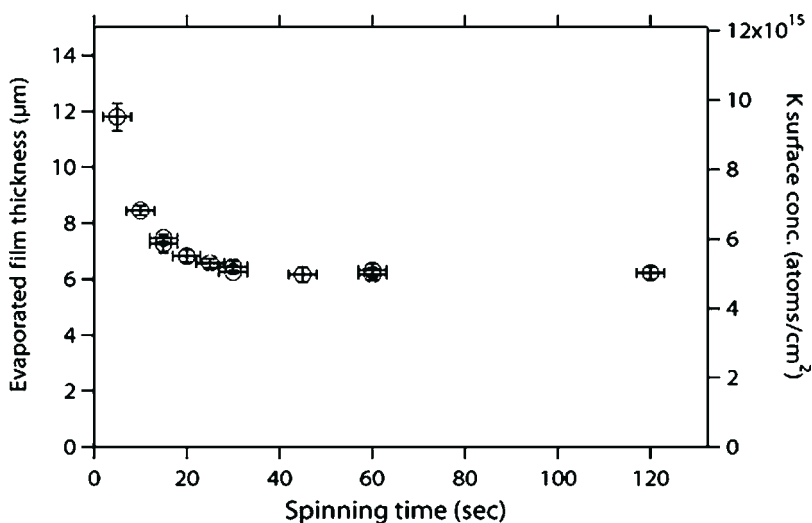


FIGURE 19.12 Evaporated film thickness calculated (using Equation (19.5)) as a function of spinning time from K surface concentration after spinning a 1000 ppm KCl solution at 500 rpm. The right axis refers to the as-measured surface concentrations (in atoms/cm²). Each marker represents an average of 13 measurements.

needed for complete evaporation, we can state that $t_{\text{dry}} \approx 45$ s. Therefore, the time between $t = 15$ and 45 s represents the evaporation phase as indicated in Figure 19.7. During this phase, the water from the film on the wafer surface is evaporating, but no salt is removed from the wafer surface. This is also confirmed by the fact that for $t > 15$ s, δ_{evap} is nearly constant. For longer times ($t > 45$ s) the film has completely evaporated and no further changes take place.

19.4.1.2 Uniformity of the Evaporated Film

The vertical error bars plotted in Figure 19.12 indicate that the uniformity of the spin drying process is on the order of 10% or better. This is in good agreement with the predictions made in Section 19.2.3.1. There it was discussed that at low Reynolds number the mass transfer coefficient for a rotating substrate (and hence the evaporation for a spin-drying process) is independent of the position on the substrate. However, it was also mentioned that deviations from this ideal behavior could be related to the acceleration of the spinner and to the presence of a splash guard that influences the gas flow. The overall effect of these parameters is investigated by comparing the results of metal salt tracer tests performed on the two spinning tools that are described in Section 19.3.1.1. In these tests, a 1000 ppm KCl solution is dispensed on the wafers followed by spin drying at 300 or 500 rpm. This value is low enough to ensure laminar flow over the entire wafer surface. The K surface concentration is measured on various positions

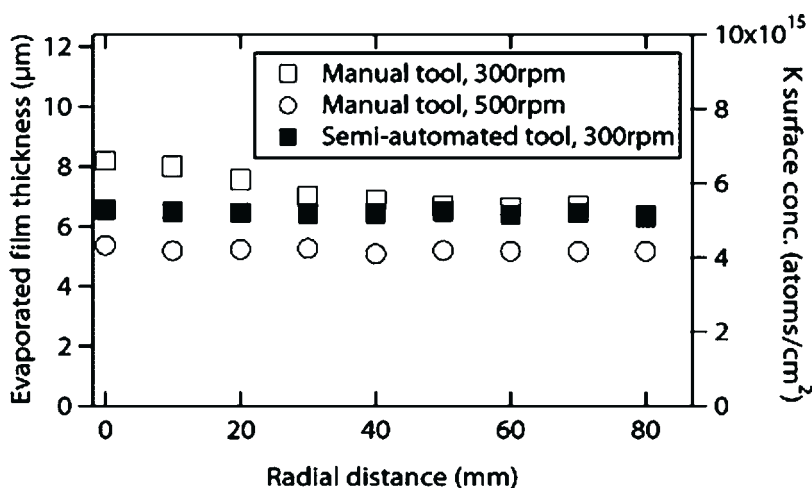


FIGURE 19.13 Evaporated film thickness (calculated using Eq. (19.5)) based on K surface concentration at various distances from the center of the wafer following a spin process after dispensing 1000 ppm KCl. The right axis represents the measured surface concentration. Symbols ($\square$), ($\circ$): manual spinner; ($\blacksquare$): semi-automatic spinner.



along the same diagonal line. A summary of the results is given in Figure 19.13 with the K surface concentration on the right axis and the corresponding evaporated thickness (calculated from Eq. (19.5)) on the left axis.

Markers of the same color represent data measured on the same TXRF tool, under the same measurement conditions. It can be seen that for the 300 rpm test on the manual spinner the values within 2 cm from the center are somewhat higher than on the semi-automatic spinner. At larger radial distances, both values agree very well. This can be attributed to the slower acceleration of the first tool. At 500 rpm the uniformity is significantly improved. This means that the average of a center point and several off-center points is a good estimate of the average wafer surface concentration. It is important to mention, however, that in some cases it was observed that the center region (with a radius of approximately 0.5 cm) yielded a concentration up to 80% lower than the off-center locations.¹¹¹ One hypothesis to explain this phenomenon is that it is related to wettability issues of the thinning liquid film. This is based on the fact that the phenomenon was observed only when acidified solutions were used. From the literature it is known that the wettability of a liquid with a silica surface is a function of the pH of the solution.⁴⁸ Because the data reported in this work are always average values obtained from measurements taken at various positions on the wafers, the contribution of the measurements taken at the center region is very small and is averaged out by the off-center measurements.

19.4.1.3 Effect of the Rotation Speed on Evaporation

Figures 19.12 and 19.13 show that for rotation speeds on the order of a few 100 rpm, the evaporated film thickness is 4–6 μm . Since evaporation of liquid results in deposition of dissolved and suspended non-volatile species, it is important to minimize the amount of liquid evaporation. According to Equation (19.20), the amount of salt residue, and hence also the amount of evaporating liquid, is inversely proportional to the square root of the rotation speed:

$$\delta_{\text{evap}} = \alpha \times \frac{1}{\sqrt{\omega}} \tag{19.31}$$

with α a fitting parameter. Equation (19.31) is fitted to experimental data obtained from metal salt tracer tests at various rotation speeds which yields $\alpha = 41.4 \times 10^{-6} \pm 0.8 \times 10^{-6}$ with all values in SI units (ω in rad/s and δ_{evap} in m). The results of these tests are summarized in Figure 19.14.

In these tests, wafers are pre-wetted with a 1000 ppm KCl solution, spun dry, and measured by TXRF. The left axis represents the evaporated film thickness calculated using Eq. (19.5). The right axis is the K surface concentration measured by TXRF. Each marker represents an average of 9 or 17 measurements performed on one or two lines across the wafer surface. The two types of markers on this figure correspond to the two spinning tools used in the experiments. Markers of the same color represent data measured on the same TXRF tool, under the same measurement conditions. The good agreement between the experimental data and Eq. (19.31) shows that entrainment of water vapor in the passing flow of gas is the dominant factor determining the final film thickness.

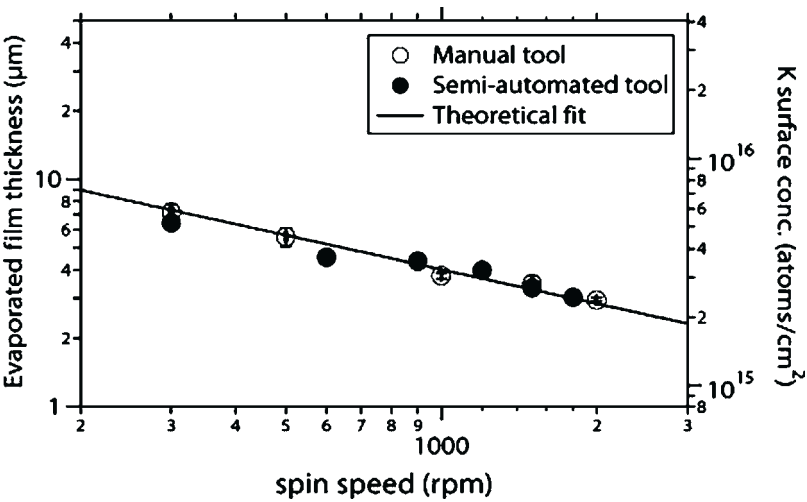


FIGURE 19.14 Evaporated film thickness calculated from the K surface using Eq. (19.5) following a spin process after dispensing 1000 ppm KCl. The right axis represents the measured surface concentration. The solid line is calculated from Eq. (19.31) with $\alpha = 41.4 \times 10^{-6} \pm 0.8 \times 10^{-6}$.

From the graph one can also see that at the highest rotation speed used in these tests (approximately 2000 rpm, which is comparable to typical spinning speeds used in the semiconductor industry) δ_{evap} is on the order of 2–4 μm . Furthermore, the height of the vertical error bars indicates that the evaporation process occurs very uniformly, even up to the highest rotation speeds tested.

19.4.1.4 Investigation of Adsorption

Once the amount of liquid evaporation is known, one can also use the spinning process in order to study adsorption of metal ions by using Eq. (19.3) to interpret the experimental results of metal salt tracer tests. Because the metal ion adsorption is influenced by the pH of the solution (see Section 19.2.2), we can distinguish adsorption from evaporative deposition by varying the pH of the solution. Evidently this assumes that the spinning process is not (or much less than adsorption) influenced by a change in liquid pH. This is evidenced in Figure 19.15 which summarizes the results of a series of 1000 ppm KCl tracer tests performed at 1800 rpm.

The pH of the solution was adjusted by adding HCl. Each marker represents an average of five measurement points taken at various positions on the wafer surface. The evaporated film thickness (calculated using Eq. (19.5)) is shown on the left axis and the K surface concentration is shown on the right axis. Under these experimental conditions, evaporation is dominant over adsorption (because the surface concentration is significantly higher than 10^{13} atoms/cm²).

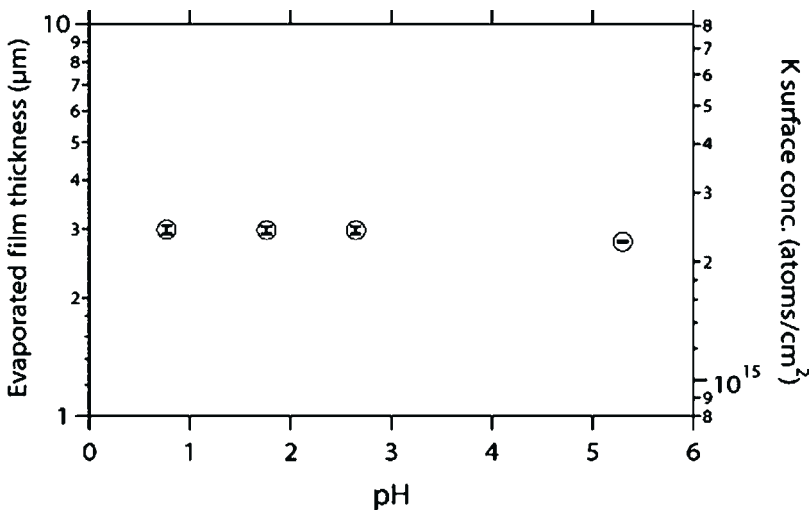


FIGURE 19.15 Evaporated film thickness calculated from the K surface using Eq. (19.5) following a 1800 rpm spin process after dispensing 1000 ppm KCl. The pH of the solution is adjusted by adding HCl. The right axis represents the measured surface concentration. Each marker represents an average of five measurement points.

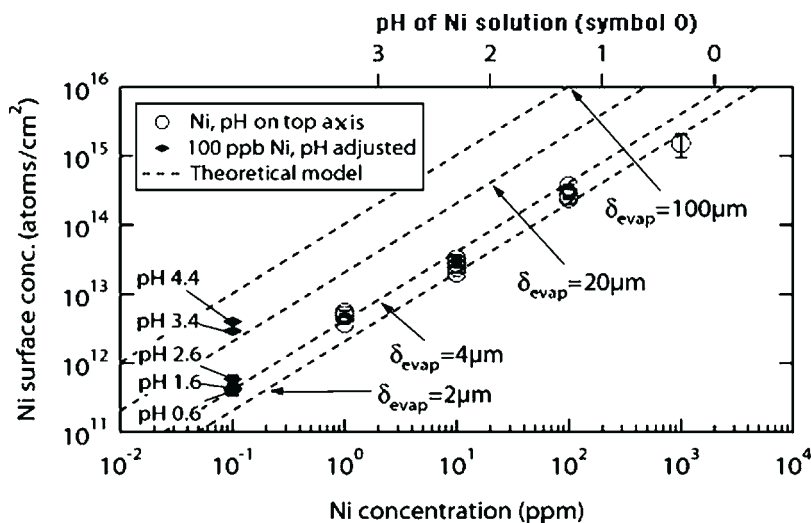


FIGURE 19.16 Nickel surface concentration as function of Ni bulk concentration in solution for a wafer pre-wetted by a Ni solution and spun at 1800 rpm. (○): pH of the Ni solution denoted on the top axis; (◆): pH values of the 1000 ppb Ni solution indicated next to marker. Dashed lines are calculations of δ_{evap} using Eq. (19.5).

Figure 19.15 clearly demonstrates that the addition of HCl does not have a significant effect on the liquid evaporation during spin drying. To study adsorption, we must lower the salt concentration such that evaporative deposition becomes very small. This is illustrated in Figure 19.16, which summarizes the results obtained from Ni metal tracer tests performed at various concentrations. Markers of the same color represent data measured on the same TXRF tool, under the same measurement conditions.

The Ni salt solutions were prepared by diluting a 1000 ppm AAS solution, stabilized with 0.5 N HNO_3 . The dashed lines represent the surface concentration neglecting adsorption (i.e. Eq. (19.5)) for various values of δ_{evap} (1, 10, 100, and 1000 μm). It is important to note the agreement between the experimental data points plotted in Figure 19.16 and the qualitative model presented in Figure 19.9. For surface concentrations above 10^{13} atoms/cm² (indicated by the open markers, with the corresponding pH denoted on the upper axis), we observe a linear relationship between bulk and surface concentrations. This indicates that evaporation is the dominant contributor to the measured surface concentration. At lower surface concentrations (indicated by the filled markers), adsorption is evidenced by the pH dependence of the surface concentration. These data points are obtained by adjusting the pH of a 100 ppb $\text{Ni}(\text{NO}_3)_2$ solution with HCl. The pH of the solution is indicated next to the marker. It can be seen that for pH values below 3, the surface concentration drops significantly as the pH is lowered. Furthermore, the values obtained

at very low pH tend toward the theoretical curve calculated for $\delta_{\text{evap}} \approx 4 \mu\text{m}$. This can be an indication that Ni adsorption is suppressed to the extent that evaporative deposition of Ni becomes the dominant mechanism again.

19.4.2 Vertical Marangoni-Based Drying

19.4.2.1 Histogram of Salt Test Results

Visual inspection of the drying process reveals that at higher drying speeds or lower gas flow rates, wet spots remain on the wafer after it has been withdrawn from the salt solution. These wet spots occur most frequently at the left and right edges of the wafer and at the bottom part close to the wafer holder. The first two cases are attributed to a too limited nozzle size, while the latter is a result of droplets clinging to the cavities in the wafer holder. Although both effects can be reduced with an improved tool design, their impact on the measured performance for the current setup needs to be taken into account properly. For this purpose, Figure 19.17 plots a histogram of all measurements performed on the vertical drying setup (converted to δ_{evap} values using Eq. (19.5)).

It can be seen that the majority of values are confined to a small region ($\delta_{\text{evap}} < 0.05 \mu\text{m}$), while a limited number of points are more or less evenly spread out between $\delta_{\text{evap}} = 0.05$ and $10 \mu\text{m}$. The first type of values are attributed to the dry regions (and will be referred to as “dry” points) while the latter are attributed to the wet spots described above. They will be referred to as “wet” points. Because of the significant difference (up to 2 orders of

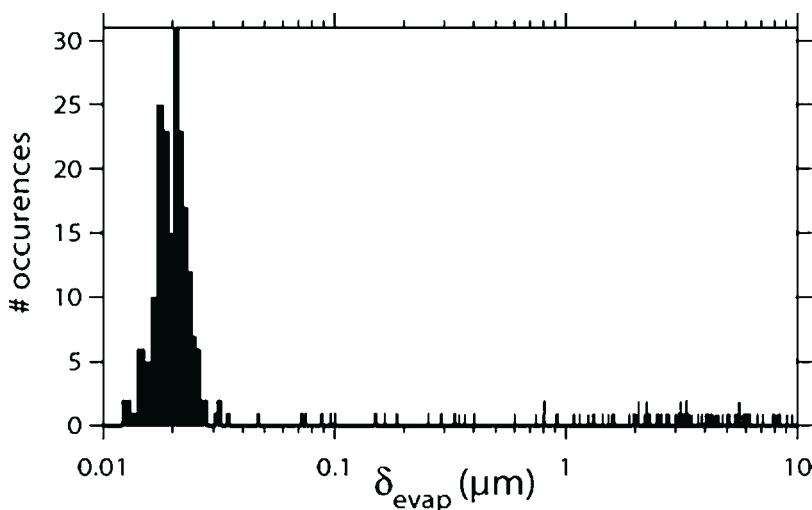


FIGURE 19.17 Histogram of δ_{evap} values (calculated from Eq. (19.5)) based on K surface concentration after withdrawal from an overflowing 1000 ppm KCl solution at various conditions.

magnitude) between the values of the dry and wet points they must be treated separately.

19.4.2.2 Effect of IPA Flow Rate

In the following series of tests, the effect of the flow rate of IPA through the nozzles has been investigated. In these tests, wafers are lifted at 7.5 mm/s from an overflowing KCl solution (≈ 1 l/ per min, 1000 ppm) at various flow rates of N_2 /IPA. The position of the delivery nozzles remains unchanged throughout the entire series of tests. The results are summarized in Figure 19.18 in terms of surface concentration for the K atoms (right axis) and the corresponding evaporated thickness calculated using Equation (19.5) on the left axis.

In this figure, the open and filled markers represent the average K surface concentration and corresponding calculated evaporated thickness for the wet and dry spots, respectively. Included in parentheses are the number of wet points and the total number of measurement points per wafer. It can be seen that the fraction of wet points decreases as the flow rate increases. This suggests that the drying performance gets better as more IPA is delivered to the surface. Furthermore, it is also observed that the average evaporated film thickness calculated using only the dry points is fairly constant (of the order of 0.04 μm) and does not depend on the fraction of wet data points. As the average surface concentration of K atoms is of the order of 10^{13} atoms/ cm^2 the corresponding

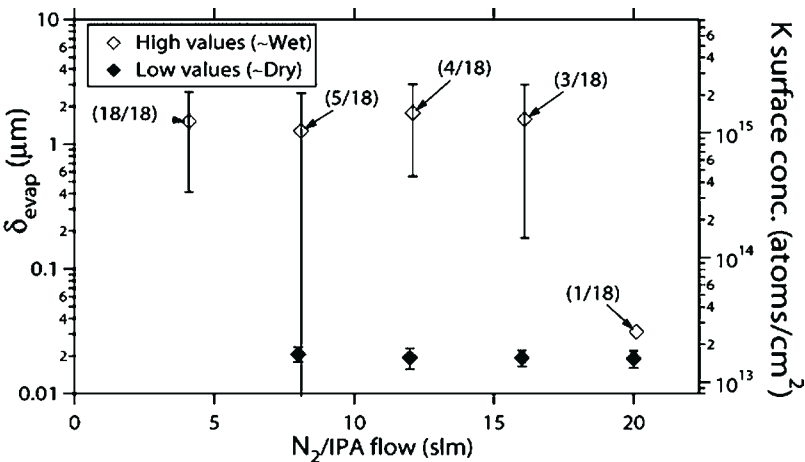


FIGURE 19.18 Evaporated film thickness (calculated using Eq. (19.5)) based on K surface concentration after withdrawal from an overflowing 1000 ppm KCl solution at 7.5 mm/sec. The N_2 /IPA flow varied between 4 and 20 slm (standard liters per minute). The open markers represent an average of all wet spots, while the solid markers represent the average of the dry spots. The numbers in parentheses are the number of the wet spots and the total number of data points for each experimental condition.

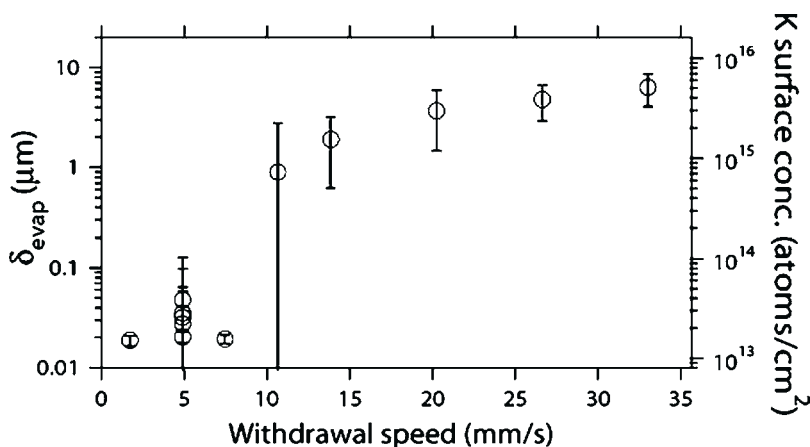


FIGURE 19.19 Evaporated film thickness (calculated using Eq. (19.5)) based on K surface concentration after withdrawal from an overflowing 1000 ppm KCl solution at various speeds. The N_2/IPA flow was 20 slm in all tests. Each marker represents an average value of several measurements performed on the same wafer.

evaporated film thickness could also be attributed to adsorption, hence the calculated value of δ_{evap} should be considered as an upper limit.

19.4.2.3 Effect of Withdrawal Speed

Next, the effect of withdrawal speed on drying performance was investigated. In these tests, wafers are lifted from an overflowing KCl solution (≈ 1 per min, 1000 ppm) at various withdrawal speeds, keeping the N_2/IPA flow constant at 20 slm (the maximum value of the mass flow controller). The results are summarized in Figure 19.19. Each marker represents an average of 18 points, measured across two lines, parallel and perpendicular to the direction of motion.

The figure shows that for withdrawal speeds up to 8 mm/s $\delta_{\text{evap}} < 0.04 \mu\text{m}$. This value agrees well with the results of the dry spots shown in Figure 19.18. Hence, in this case also the results can be attributed to adsorption. A reproducibility study (using five wafers in identical conditions at 5 mm/s) indicates good agreement between the different values, although in some of the tests a wet spot could be observed. As discussed in Section 19.4.2.1, these wet spots were often found at the edge of the wafer and are attributed to an insufficient delivery of N_2/IPA gas by the nozzle at the edges of the wafer. Figure 19.19 also shows that when the withdrawal speed exceeds 8 mm/s a very significant increase in the salt concentration is observed. This is in agreement with visual observations that show that when the withdrawal speed exceeds 8 mm/s, the meniscus region (i.e., the transition zone between the dry (withdrawn) and the wet (immersed) part of the wafer) becomes unstable. Under these conditions, one clearly sees

that liquid is withdrawn with the wafer from the bath, resulting in liquid covering almost the entire wafer surface.

19.4.3 Marangoni-Based Drying on a Rotating Wafer

19.4.3.1 Effect of Liquid Dispensation During Drying

The previous data have shown that for good drying performance, the meniscus region must remain stable throughout the process. To investigate whether similar reasoning applies to the rotational Marangoni process, a pre-wetted wafer was dried using only the N₂/IPA delivery nozzle. No liquid was dispensed during the drying step.¹¹² During the tests, it was observed that thick rims were formed in the water film. These rims were rapidly spun off the wafer. This phenomenon is, however, not limited only to the region underneath the translating IPA nozzle, but it also occurs over the entire water film that covers the wafer surface. This makes the process very uncontrollable. The results of this test, using a 1000 ppm KCl solution and a rotation speed of 300 rpm, are summarized in Figure 19.20.

When comparing the evaporated thickness with the result for a spin drying process at 300 rpm ($\approx 8\text{ }\mu\text{m}$, see Figures 19.13 and 19.14), it is observed to decrease by at least a factor of 2. In agreement with visual observations, the resulting film thickness is very non-uniform across the wafer diameter. Furthermore, the values obtained are still more than one order of magnitude larger than obtained with the vertical Marangoni-based drying process. This proves the

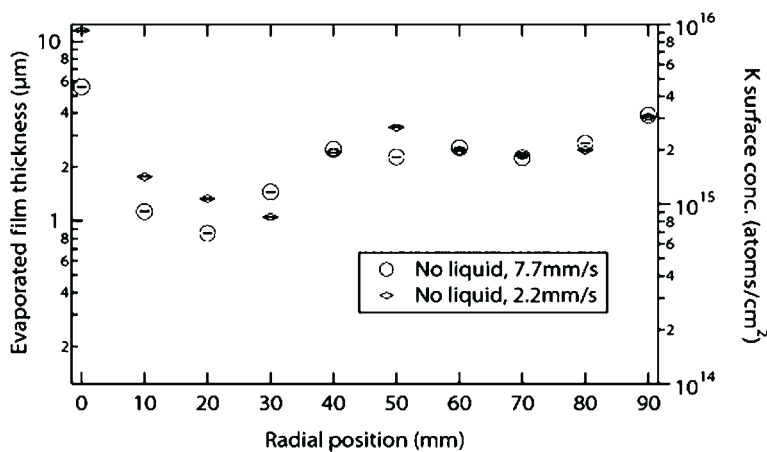


FIGURE 19.20 Evaporated film thickness (calculated using Eq. (19.5)) based on K surface concentration (right axis) of a wafer dried with the rotational Marangoni drying process operated at 300 rpm without dispensing liquid during the drying step. The salt solution used to pre-wet the wafer was a 1000 ppm KCl solution.

need for additional water dispensation during the drying process to maintain control over the meniscus region as the nozzles move over the wafer surface.

19.4.3.2 Effect of Nozzle Speed

One important distinction between Marangoni-based drying on a vertically moving wafer and on a rotating wafer is that the fluid dynamic boundary conditions are similar on all points of the vertically moving wafer. This is not the case for rotational Marangoni-based drying since some of the process variables depend on the radial position on the wafer, such as the centrifugal force acting on the liquid. In Section 19.2.3.3, it was shown that rotational Marangoni-based drying would exhibit the same qualitative trends as Marangoni-based drying on a rotating wafer provided a number of assumptions are valid. This implies that for a constant mean surface tension gradient $\bar{\tau}$, the rotational Marangoni-based process is independent of the radial position. To verify if this is indeed the case, the results of a series of metal salt tracer tests are shown in Figure 19.21.

In these tracer tests (also described in more detail in Section 19.3.4.3) a 1000 ppm KCl solution was delivered through the second nozzle. The wafer rotation speed was set at 300 rpm. Various nozzle speeds were tested: 1, 3.7, 5, and 12.5 mm/s. The tests were performed on the two different platforms described in Section 19.3.1.3. Markers of the same color represent data measured on the same TXRF tool, under the same measurement conditions. It is important to realize that, besides the nozzle speed and wafer rotation speed, there are many tool settings that are not accurately known or that can vary from experiment to experiment, such as the nozzle height, nozzle direction, and liquid impact speed. These parameters can result in tool-to-tool

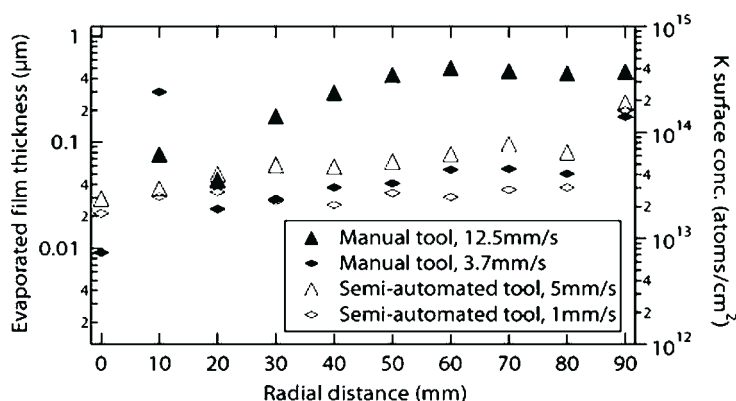


FIGURE 19.21 Evaporated film thickness (calculated using Eq. (19.5)) based on K surface concentration in a rotational Marangoni test with a 1000 ppm KCl solution at 300 rpm for various nozzle speeds. The corresponding K surface concentration is shown on the right axis. Each marker represents 1 TXRF measurement.

differences which were not investigated in detail in this study. The results from Figure 19.21 indicate that in all cases the signal increases from the center to the edge of the wafer. For the lowest nozzle speeds (<5 mm/s) the increase is, however, rather small (less than a factor of 2). This means that the assumptions from Section 19.2.3.3 are not too unrealistic. For higher nozzle speeds (>10 mm/s), the increase from the center to the edge is larger, nearly one order of magnitude. This suggests that the surface tension gradient induced between the adjacent liquid and tension active vapor flows decreases with the distance to the wafer center, and that the assumptions made in Section 19.2.3.3 are no longer valid.

The large deviations in the data points close to the center are related to the formation of the first dry spot at the beginning of the drying process. Especially in the case of the manual setup, it is sometimes observed that in the wafer center a small droplet is formed which then rapidly evaporates. Also at the edge of the wafers the surface concentration increases rapidly. This is because in the setups used in this work, in some cases an amount of liquid was left on the wafer bevel when the nozzles reached the edge of the wafer. When no additional measures are taken, this liquid can spread out on the wafer surface and evaporate, resulting in a locally high surface concentration of metal salt.

From a qualitative comparison of all curves, we can also observe that the average surface concentration increases with the nozzle speed. For the lowest speed tested (1 mm/s) we find a result which is rather comparable to the vertical Marangoni drying process. Because this value is on the order of 10^{13} atoms/cm², adsorption may be the dominant effect. The calculated value of $\delta_{\text{evap}} = 0.02\text{--}0.03$ μm must therefore be considered as an upper limit. An increase in the nozzle speed to 5 mm/s results in a twofold increase in the surface concentration. Because this value is significantly higher than 10^{13} atoms/cm², it indicates that the drying process is unable to remove all the liquid in the time allowed by the moving nozzles. For higher speeds (above 10 mm/s), the resulting surface concentration increases by more than a decade. Remarkably, these findings are similar to what was found on the tests with the vertical Marangoni drying unit (see Figure 19.19).

19.4.3.3 Effect of Liquid Surface Tension

As discussed in Section 19.1.3.2, the driving force for Marangoni drying is the difference in surface tension across the meniscus region which is caused by a concentration difference. The effect of the IPA content on the surface tension of an IPA/water mixture is given in Figure 19.22.

It can be seen that small amounts of IPA result in a large decrease in the surface tension of water, which explains the strong Marangoni effect that is observed. For the IPA/water system, the maximum surface tension difference is approximately 51 mN/m, based on values of 72 and 21 mN/m as the surface tensions of pure water and IPA, respectively. To investigate the sensitivity of

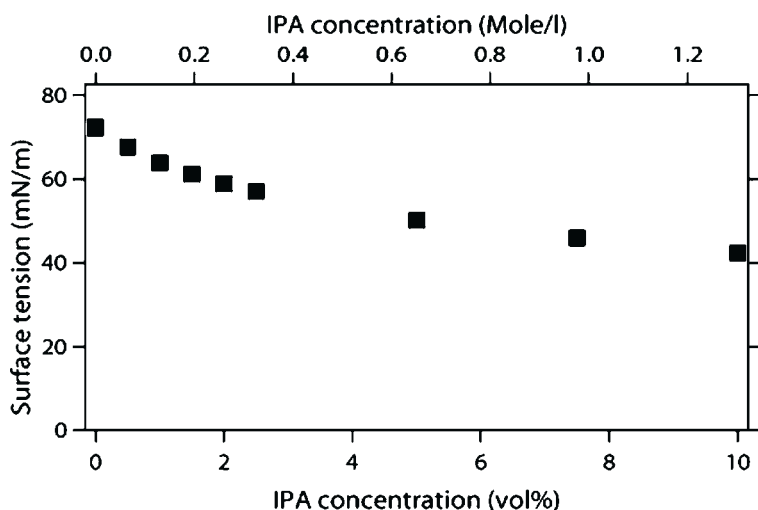


FIGURE 19.22 Surface tension of IPA/H₂O mixtures for various IPA concentrations determined using the Wilhelmy plate method.

the Marangoni drying process to the maximum surface tension difference, a series of metal salt tracer tests was performed where a mixture of water, liquid IPA, and metal salts was used as a tracer solution. Because of the addition of IPA, the surface tension of this tracer liquid was lower than that of a solution based on pure water. The results of these tests are summarized in Figure 19.23. The tracer solution used was a 1000 ppm KCl solution containing 0, 1, 5, or

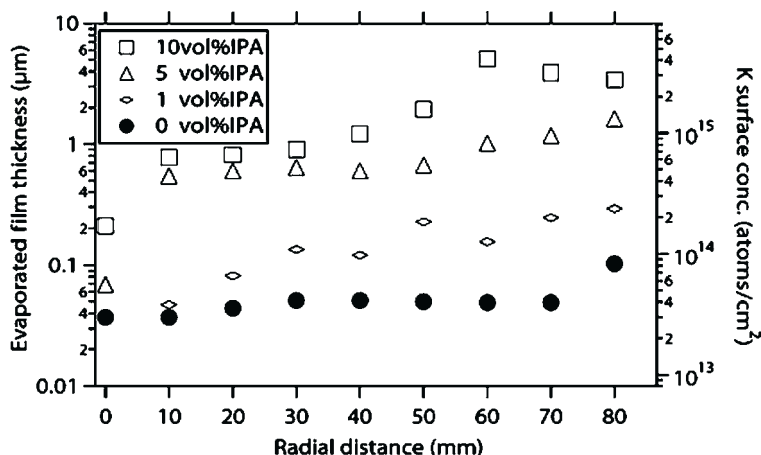


FIGURE 19.23 Evaporated film thickness (calculated from Eq. (19.5)) using K surface concentration given on the right axis) for a rotational Marangoni drying process operated at 300 rpm and 5 mm/s with a 1000 ppm KCl solution containing 0, 1, 5, and 10 vol% IPA.

10 vol% IPA. The tests were performed on the semi-automated tool at a wafer rotation speed of 300 rpm and nozzle speed of 5 mm/s.

Figure 19.23 shows that even a small decrease in liquid surface tension from 72 to 64 mN/m for 1 vol% IPA results in a significant decrease in drying performance. Because the largest decrease in surface tension occurs for the highest IPA concentrations, more IPA is needed in the meniscus region to obtain the same driving force (total surface tension difference with respect to the bulk liquid). For the same flux of IPA to the meniscus region, therefore the Marangoni effect is expected to reduce as the bulk liquid surface tension decreases. This is evidenced by the experimental results. Additionally we can also observe in Figure 19.23 that for all cases where IPA is added to the tracer solution the K signal increases toward the edge of the wafer.

19.4.3.4 Investigation of Adsorption

Since the surface concentrations measured in the metal salt tracer test for Marangoni drying are on the order of 10^{13} atoms/cm², we cannot distinguish between adsorption and evaporation. In Section 19.4.1.4, it was shown that this distinction could be made by performing metal salt tracer tests with varying liquid pH and metal salt concentration. The results of such a series of tests (which are described in more detail in Section 19.3.4.2) are summarized in Figure 19.24.

The tests were performed on the semi-automated tool at a rotation speed of 300 rpm and a nozzle speed of 1 mm/s. The open markers represent dilutions of a 1000 ppm Ni(NO₃)₂ AAS standard solution containing 0.5 N HNO₃ in water. The corresponding pH is given on the top axis. For the 100 ppb solution the pH

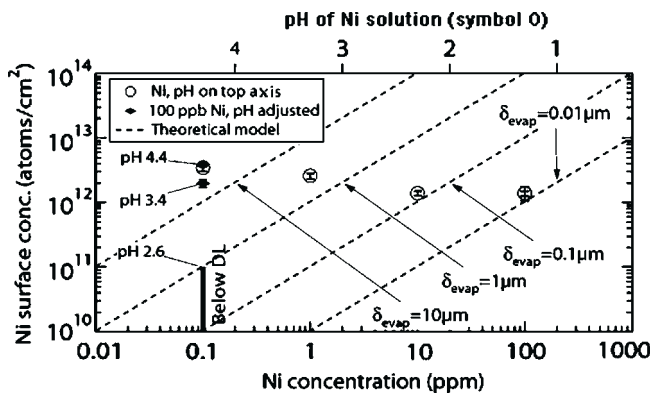


FIGURE 19.24 Surface concentration of nickel after rotational Marangoni drying of wafer pre-wetted with Ni salt solution. The process was performed on the semi-automated tool at 300 rpm with a nozzle speed of 1 mm/s. For the open markers, the pH of the solution is indicated on the top axis; for the solid markers, the pH values are listed next to the marker. Dashed lines represent Equation (19.5) for various values of δ_{evap} .

is lowered by adding HCl. These points are denoted by the filled markers. Each of the markers represents the average of four TXRF measurement points taken 2 cm apart, starting from the center of the wafer and moving radially outward. Markers of the same color are measured on the same TXRF tool under the same measurement conditions. The dashed lines are calculated from Equation (19.5) for various values of δ_{evap} . Unlike the results in Figure 19.16, we do not observe a region where the Ni bulk concentration is proportional to the Ni surface concentration. This indicates that the corresponding surface concentrations are either due to adsorption or due to the formation of an equilibrium film thickness determined by surface forces. The fact that lowering the pH reduces the surface concentration is a strong indication that one is observing the result of an adsorption process, although an effect of the HCl addition on the drying process cannot be ruled out. To estimate the evaporated film thickness, we can use the right most experimental data point as an upper limit. This value is on the order of 20 nm which is similar to the results obtained with KCl salts. It should be emphasized that salt tests are not ideal to study the evaporated film thickness for a Marangoni drying process because the addition of salts can also influence the drying process. This is because for a Marangoni drying process, the corresponding evaporated film thickness is so small that surface forces become dominant (see Section 19.2.1).

19.5 DISCUSSION OF EXPERIMENTAL DATA

19.5.1 Spin Drying

19.5.1.1 Spin-Off vs. Evaporation

First, it is very important to realize that δ_{evap} and h_{final} as plotted in Figure 19.7 represent two different values. The term h_{final} , on the one hand, corresponds to the (normalized) physical thickness of the salt residue, assuming it forms a uniform deposit. The term δ_{evap} , on the other hand, represents the thickness of the liquid film from which this solid residue is left behind upon evaporation of the liquid. These two values are related by

$$\delta_{\text{evap}} = h_{\text{final}} F \frac{\rho N_A}{CM_W} \quad (19.32)$$

where M_W is the salt molecular weight (kg/mole), N_A is Avogadro's number, and ρ is the salt density (kg/m³). Because δ_{evap} is proportional to h_{final} , we can use the same trends derived in Section 19.2.3.1 for h_{final} to calculate δ_{evap} using Eqs. (19.15) and (19.20). For the experimental conditions of the salt tracer tests (see Section 19.3.4) we have:

$$\frac{\rho N_A}{CM_W} \gg 1 \quad (19.33)$$

and

$$\delta_{\text{evap}} \gg h_{\text{final}} \quad (19.34)$$

Thus, from Figure 19.7, we can see that

$$\delta_{\text{evap}} \approx h_{\text{spin-off}}(t > t_{\text{spin-off}}) \quad (19.35)$$

This means that the value of δ_{evap} obtained in the experiments represents the amount of liquid that is left on the wafer after the spin-off phase which evaporates during the evaporation phase. The results indicate that during a typical spin drying process a film of several micrometers thickness evaporates. It is interesting to note that a similar value for the evaporated thickness was obtained for tests where small particles were used instead of metal salts.¹¹³ This proves indeed that the evaporative deposition mechanism is independent of the type of tracer used as long as the size of this tracer does not influence the evaporation process. Furthermore, this is also direct evidence that the drying process can be responsible for the deposition of particles on a surface.

Furthermore, the data from this work also show that the evaporated film thickness is inversely proportional to the square root of the spin speed. From the models describing photoresist coating, we learn that entrainment of liquid in the gas phase is the limiting transport factor in the evaporation process. Using Eq. (19.21), we can calculate that the flow rate of entrained gas is approximately 60 slm for a 200 mm wafer rotating at 1000 rpm (assuming $\nu_g = 14.8 \times 10^{-6} \text{ m}^2/\text{s}$). Because the duration of the spin-off phase is very short compared to the total process time for drying, the spinning process can also be used as a means for layer thinning. In industrial spray tools, a short intermediate spinning step is sometimes used to reduce the liquid film thickness in between two process steps or before a rinsing step is started.¹¹⁴

19.5.1.2 Effect of High Rotation Speeds

As shown in Table 19.3, increasing the rotation speed to reduce the evaporated film thickness has the advantage of higher throughput (by lowering the total drying time), but too high a rotation speed can result in a system with turbulent gas phase conditions. If this is the case, evaporation will be higher than predicted by the model in Section 19.2.3.1 because in the turbulent regime, mass transfer will be higher than under laminar conditions.^{12,85,92} In the experimental conditions tested in this work ($\omega < 3000 \text{ rpm}$), no evidence of a turbulent regime was observed. An additional disadvantage of a high spinning speed is the entrainment of contaminants from the ambient atmosphere due to the gas flow. Furthermore, higher spinning speeds also increase the risk of splash back when droplets that are rapidly spun outwards are insufficiently removed by the catch cup around the spinner and can end up on the wafer surface where they evaporate and result in surface defects.^{11,12} One important point to stress is that the effects mentioned above are all expected to become more important as the wafer size increases. Not only the Reynolds number at the edge, but also the entrained

air flow will increase with the wafer size resulting in more evaporation and more entrained particles. Furthermore, the circumferential velocity of the wafer increases, which requires a more complicated tool design and higher exhaust flows to minimize the risk of splash back.

19.5.1.3 Wafer Topography and Surface Heterogeneity

Under real processing conditions, only a very small fraction of the substrates processed will be covered with a homogeneous and uniform film similar to the substrates tested in this work. In most cases, a typical wafer surface contains a specific pattern that can consist of adjacent areas of different materials or areas with significant height differences. These heterogeneities will have a serious impact on the performance of the drying process. In the last stages of a spin drying process, there will be only evaporation, and almost no lateral flow in the liquid film. The main factor determining film breakup is expected to be the local surface wettability. It has indeed been observed from salt tests⁵ that on a surface consisting of areas of different wettabilities, the film can break up at the hydrophobic regions and collect on the adjacent hydrophilic areas. It must be realized that this phenomenon is very sensitive to the relative areas of hydrophilic and hydrophobic surfaces. If the more hydrophobic areas are significantly larger than the typical length scale of film stability (of the order of 100 nm), the liquid film can break up into micro-droplets which will evaporate locally. This effect has been observed in tests with nitrogen blow drying on copper damascene structures and on shallow trench isolation structures.^{5,115} Because the boundary layer for evaporation is several hundred micrometers thick (calculated from Eq. (19.18)), the evaporation process takes place over a relatively large length scale and is not influenced by the substrate topography. However, there is evidence that on a local scale micro-droplets can still move on the substrate under the influence of centrifugal forces and become trapped at topographic height differences.^{5,115,116} Furthermore, topography will create recessed volumes (“trenches”) in which all residues of the evaporating liquid can become collected. It was shown, however, that even in the case of trenches with a very high aspect ratio, the amount of residue had no significant effect on device performance when liquids with a purity in the ppb range were used.¹¹⁷

19.5.2 Marangoni-Based Drying

19.5.2.1 Limits to the Use of Salt Tests

The salt tracer tests, as described in this chapter, have some limitations when they are used to investigate a Marangoni-based drying process. Under the conditions used in this work, a distinction between evaporation and adsorption could not be made. One problem is the uncertainty in the adsorption process (more specifically in the number of possible exchange sites to which the metal salt ions can bind). Second, the evaporated film thickness predicted by the salt

tests is of the order of only a few tens of nanometers. This means that during the Marangoni-based drying process, the liquid film is thinned down to the point where its stability is mainly governed by surface forces (as described in [Section 19.2.1](#)). In this respect, it is important to realize that by adding salts to a liquid the electrostatic and van der Waals interactions are screened as a result of an increased ionic strength. Moreover, the salt addition can also influence short range repulsive structural forces.^{43,48} Another limitation of the salt tracer tests is related to the fact that the surface charge density of silica is also known to decrease in alcohol/water mixtures as the concentration of alcohol increases⁶⁴; hence, the interaction between the cations and the surface may be influenced by the use of the IPA during the Marangoni-based drying process. Furthermore, the solubility of the salt tracer is also a function of the amount of tensioactive component. For example, for the case of KCl, the solubility in alcohol is lower than in water.¹¹⁸ Thus, if the local concentration of IPA exceeds a certain value, a certain amount of KCl could precipitate on the surface. However, as the concentration profile of IPA in the liquid meniscus region is not known, the extent of this phenomenon cannot be assessed. Due to these limitations one can only conclude that for Marangoni-based drying the salt residues measured on the surface cannot unambiguously be attributed to adsorption, precipitation, or evaporation. However, they can be used to calculate an upper limit to the amount of liquid evaporation that takes place during the process.

19.5.2.2 Mean Surface Tension Gradient for Marangoni-Based Drying

An important parameter in the Marangoni-based drying process introduced by physical modeling is the mean surface tension gradient $\bar{\tau}$ and the corresponding Marangoni number Ma . Results for both parameters are listed in [Table 19.1](#) from available literature. For the sake of comparison, we have also calculated values for these parameters using Eqs. (19.26) and (19.29) for the experiments reported in this work. The resulting values are given in [Table 19.4](#), grouped by the type of experiment and reported as the total range observed for each particular condition.

We can see that the values of $\bar{\tau}$ reported in [Table 19.4](#) are of the same order of magnitude as the results reported in [Table 19.1](#). However, for each experimental condition, the reported values show a significant spread. This means that for a given set of experimental conditions the mean surface tension gradient $\bar{\tau}$ is not a constant value, but depends on at least one of the process variables. For the case of vertical Marangoni-based drying, $\bar{\tau}$ depends on the withdrawal speed; in the case of rotational Marangoni-based drying, $\bar{\tau}$ depends on the nozzle speed as well as the radial distance in some cases. This means that the mean surface tension gradient can only be used in a qualitative manner to compare two drying techniques with each other. For a correct characterization of a given set-up, one should also make correlations between $\bar{\tau}$ and the appropriate process variables (such as IPA delivery and drying speed).

TABLE 19.4 Summary of the Mean Surface Tension Gradient $\bar{\tau}$ (Calculated from Eq. (19.29)) and the Marangoni Number Ma (Calculated from Eq. (19.26)) for the Experimental Results Obtained in this Work

Condition	Detail	U (mm/s)	Radial Distance (mm)	$\bar{\tau}$ (N/m ²)	Ma (Dimensionless)
Vertical		1.7–7.5		40.6–169.7	41.3–135.7
Vertical		10.7–33.0		5.2–2.3	3.7–1.1
300 rpm	No H ₂ O	2.2	20–80	1.1–0.4	1.3–0.4
300 rpm	No H ₂ O	7.7	20–80	2.5–1.3	2.0–1.1
300 rpm		1.0–12.5	20	12.6–121.6	19.8–82.1
300 rpm		1.0–12.5	80	11.5–12.0	18.0–8.1
300 rpm	H ₂ O (IPA0–10%)	5	20	48.9–2.6	44.7–2.4
300 rpm	H ₂ O (IPA0–10%)	5	80	20.9–0.6	19.1–0.6

Ma is a Dimensionless number.

Compared to the reported data from Marra and Huethorst³³ (summarized in Table 19.1), we can see that the experimental results obtained in this work cover a much wider process window. For vertical Marangoni-based drying as well as for rotational Marangoni-based drying, we observe good drying performance (high values of both $\bar{\tau}$ and Ma) for speeds up to 10 mm/s. Furthermore, we can also see that at even higher drying speeds the Marangoni number is still on the order of 1–5. From Figure 19.8, we can see that this results in a decrease of the entrained thickness by a factor of 2–10. Under these conditions, the wafer surface is still covered by a thin liquid layer, but due to the Marangoni effect this layer is significantly thinner than, for example, a carry-over layer obtained without Marangoni effects.¹ Even for non-optimal cases where only N/IPA is delivered and no liquid, the Marangoni number is seen to be of the order of 1, resulting in a reduction of the liquid film thickness by a factor of 2.

To check if the reported values for $\bar{\tau}$ in Tables 19.1 and 19.4 are physically realistic, we can estimate a typical value as follows: a typical length scale over which the surface tension gradient effect operates is the size of the liquid meniscus. The latter can be estimated from the so-called capillary length $\sqrt{\gamma/\rho g}$ ⁹⁸ which is of the order of 3 mm for water. Furthermore, the maximum difference in surface tension that can be obtained for water (with $\gamma \approx 72$ mN/m) and a pure tensioactive liquid like IPA (with $\gamma \approx 21$ mN/m) is approximately 51 mN/m.

Consequently, the estimated mean surface tension gradient is of the order of $15\text{--}20\text{ N/m}^2$, which is of the same order of magnitude as the values of $\bar{\tau}$ reported in [Tables 19.1](#) and [19.4](#). A comparison between the two types of gas flows (listed in [Table 19.1](#)) shows that the Marangoni effect is less pronounced in a semiquiescent atmosphere than in a directed vapor stream. This suggests that mass transfer of IPA from the gas phase to the liquid phase is a limiting factor, a finding previously observed by other researchers.^{18,33,119} This statement is also substantiated by the fact that for both types of gas flow $\bar{\tau}$ depends on the withdrawal speed instead of being constant. Other researchers, however, claim that mass transfer limitations occur in the liquid phase.³⁵ Most likely, the mass transfer limiting factor is largely dependent on the tool design, for example, by the way the tensioactive vapor is delivered to the meniscus region.

19.5.2.3 *Drying Speed*

From the point of view of drying time, it is important to realize that Marangoni-based drying is essentially mass transfer limited (see also [Section 19.5.2.2](#)). This means that for both vertical as well as rotational Marangoni-based drying increasing the wafer size will increase the drying time. For the case of vertical Marangoni-based drying, the IPA delivery system (e.g. nozzles) will scale linearly with the wafer size and so will the total drying time. This is not the case for rotational Marangoni-based drying since [Table 19.4](#) shows that for larger distances from the axis of rotation the drying performance decreases. This means that for an optimal rotational Marangoni-based drying process the nozzle speed must decrease with the distance from the center, hence the average drying speed decreases as the wafer diameter increases. The throughput can be increased by increasing the mass transfer, for example, by improving the nozzle delivery system. The main drawback of such an optimization, however, is that as mass transfer in the gas phase increases, this will also result in an increase of evaporation of liquid from the meniscus zone. Evidently, there is still room for improvement by carefully investigating and optimizing the role of each process parameter (flow rate, incidence angle, and nozzle position), but the total improvement is not expected to be an order of magnitude.

19.5.2.4 *Wafer Topography and Surface Heterogeneity*

In contrast to spin drying (see [Section 19.5.1.3](#)), the effect of surface topography and heterogeneity will be different in a Marangoni-based drying process. The reason is that the Marangoni effect causes the liquid film to attain an apparent finite contact angle on hydrophilic, as well as hydrophobic surfaces.³³ Experimental results indicate indeed a reduced number of defects associated with drying, especially on surfaces of mixed hydrophilic/hydrophobic nature.^{5,115,116} However, Marangoni-based drying is more sensitive to surface topography. This is because the dimension of the thinnest part of the meniscus region is comparable to the size of a typical topographic height difference on a semiconductor substrate. When the drying front crosses such a topographic step, the liquid film

can rupture, causing an amount of liquid to remain close to the height step. This phenomenon has been observed on tests with wafer pieces, using a manual Marangoni-like procedure,¹¹⁵ but it must be mentioned that vibrations induced by the experimental procedure can also explain this behavior. For very large topographic height differences (i.e. very deep trenches), no information is currently available, but intuitively it is expected that for deep trenches the drying front will rupture at some point, leaving an amount of liquid inside the trench to evaporate. As discussed in Section 19.5.1.3, however, this is not expected to cause serious issues. Moreover, Marangoni-based drying may offer some advantages with respect to drying of deep trenches because the risk of pattern collapse will be reduced. The latter is a mechanical failure in the material, induced by the capillary force in the trench. Since water has a high surface tension, pattern collapse can become a serious issue in future device generations. The use of surface tension reducing agents (such as in Marangoni-based drying) can provide some benefits in this case.

19.5.2.5 Alternatives to IPA as a Suitable Tensioactive Component

All results reported in this paper have been obtained using IPA as the tensioactive component in Marangoni-based drying. In a detailed study of several organic liquids, Marra and Huethorst³³ proposed IPA as one of the best candidates to be used as the tensioactive component. However, the widespread use of IPA in Marangoni-based drying is most likely related to the fact that it was already generally used as a solvent in the microelectronics industry.

The choice of an alternative product with improved drying performance is still limited to a qualitative level. This is mainly because there is no model available to describe the mass transfer in the meniscus region. Without such a model, it is not straightforward to predict the properties that have a dominant effect on drying performance. First, it has been proposed that a good tensioactive component should have a high vapor pressure and a large Henry constant in order to have a high concentration in the gas phase. Furthermore, the tensioactive component must be very soluble in the liquid in order to locally reduce the surface tension.³³ However, the reduction in surface tension depends not only on the total surface tension difference between the liquid and tensioactive material, but also on the change in surface tension with concentration $\partial\gamma/\partial C$. An additional parameter is the diffusivity of the tensioactive component in the liquid which will also have an important effect on drying performance because it is directly responsible for the formation of a concentration profile at the surface of the meniscus. Since from a chemical point of view these parameters are correlated with each other in a quite complex manner, an optimization can only be done if the effect of each parameter can be quantitatively evaluated.

19.5.2.6 Residues of Organic Species After Marangoni-Based Drying

An important consequence of the use of organic molecules in Marangoni-based drying is the fact that they can adsorb on a silicon wafer surface. APIMS

(atmospheric pressure ion mass spectrometry) measurements on wafers dried in IPA vapor have confirmed the presence of IPA molecules that desorbed from the surface after heating the wafer to 400 °C.^{120,121} Moreover, it has also been observed that the wafers treated with IPA resulted in a different configuration of ODS (octadecylsiloxane) self-assembled monolayer (SAM) islands.¹²² The authors have ascribed this phenomenon to the presence of surface ester (Si–O–R) groups of the IPA. However, it should be mentioned that the esterification between low molecular weight alcohols (such as ethanol) and surface hydroxyl groups on a silicon wafer have only been reported following a 300 °C treatment.¹²³ Because during Marangoni-based drying the concentration of organic molecules as well as the ambient temperature is rather low, a surface reaction (such as esterification) will not necessarily take place. However, the fact that temperatures of up to 400 °C are necessary to desorb IPA molecules from a silicon wafer is an indication that the IPA molecules are strongly adsorbed on the surface. This finding is in agreement with the observations of SRR interactions between alcohols and silica surfaces, as reported in the literature.⁵⁴

19.6 SUMMARY AND CONCLUSIONS

In this work, the performance of several drying techniques commonly used in the semiconductor manufacturing industry is evaluated. This is done by measuring the residues on a wafer onto which a solution containing metal salts acting as tracer elements has been dispensed and dried. To correctly interpret the experimental data, the results are compared with predictions from a theoretical model. This model assumes two distinct mechanisms for deposition: adsorption and evaporative deposition. The first mechanism is a result of attractive interactions between the contaminant and the wafer surface, while the second mechanism is due to liquid evaporation during drying. For the latter case, the evaporated film thickness (the volume of liquid evaporating per unit wafer area) is introduced as a figure of merit for the drying process under study.

In the tests, spin drying was compared with two types of Marangoni (surface tension gradient) based drying: on a vertically moving wafer and on a horizontally rotating wafer. The results show that for spin drying two consecutive phases occur: during the first seconds of spinning convective removal of liquid is the dominant mechanism, followed by a phase where evaporation takes over. This behavior is confirmed by models reported in the literature describing photoresist coating. The amount of liquid evaporating during spin drying is inversely proportional to the square root of the rotation speed. This suggests that entrainment of liquid by the gas flow over the wafer surface is the dominant mechanism for evaporation. This finding is in agreement with fluid dynamics models describing the flow of gas entrained with a rotating substrate. Metal salt tracer tests indicate that for spinning speeds between a few 100 and a few 1000 rpm, the evaporated film thickness is on the order of a few μm for water. More importantly, these

findings allow a correct re-interpretation of existing data on spin coating of metal solutions where until now the distinction between adsorption and evaporation had not been taken into account by the authors.

For surface tension gradient-based drying the evaporated film thickness determined from the metal salt tracer tests is of the order of a few tens of nanometers for water. However, it must be mentioned that for these particular conditions the value can only be considered as an upper limit. The tests also indicate that for these drying processes mass transfer is a rate limiting step, limiting the total drying speed to a few mm/s for the setups investigated in this work (the vertical Marangoni-based drying process as well as Marangoni-based drying on a rotating wafer). Using a theoretical model from the literature, a mean surface tension gradient $\bar{\tau}$ is calculated to quantify the efficiency of the Marangoni effect. It is found that this value is not a simple constant for a given setup, but is a complex function of many process variables. By comparing the values of $\bar{\tau}$ calculated for the experiments performed in this work with previously reported data, it is seen that they are of the same order of magnitude but that the experiments from this work cover a much wider process window.

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